FANUC MT-LINK*i*

Database Schema

NOTE

1 This manual describes the operation of WebUI.
2 The following table lists the manuals related to MT-LINK*i*. These manuals are bundled into the DVD.
This manual is indicated by an asterisk (*).

Manual Title	Contents	Specification Number	
FANUC MT-LINK*i*			
Startup MANUAL	This manual describes basic functions and operation procedures.	A-40827 -00001EN	
Install MANUAL	This manual describes how to install, update and uninstall MT-LINK*i*.	A-40828 -00001EN	
AdminTool OPERATOR'S MANUAL	This manual describes the operation of FANUC MT-LINK*i* AdminTool (system administration software).	A-40829 -00001EN	
WebUI OPERATOR'S MANUAL	This manual describes the operation of WebUI.	A-40830 -00001EN	
Hardware Key Management Tools OPERATOR'S MANUAL	This manual describes the operation of Hardware Key Management Tools.	A-40871 -00001EN	
Database Schema	This manual describes the data configuration of the database in both MT-LINK*i* and MT-LINK*i* Integration Server.	A-40950 -00001EN	*
DBMigrationTool MANUAL	This manual describes how to use FANUC MT-LINK*i* DBMigrationTool.	A-41012 -00001EN	
Expanding Database Guide	This manual describes how to expand the databases for MT-LINK*i* and MT-LINK*i* Integration Server by using external storage.	A-41621 -00001EN	
Custom WebUI MANUAL	This manual describes how to create Custom WebUI.	A-42146 -00066EN	
Troubleshooting MANUAL	This manual describes how to address troubles which have occurred while running MT-LINK*i* / Integration Server system.	A-42146 -00187EN	
FANUC MT-LINK*i* Integration Server			
Integration Server Startup MANUAL	This manual describes basic functions and operation procedures of MT-LINK*i* Integration Server.	A-41086 -00001EN	
Integration Server Install MANUAL	This manual describes how to install, update and uninstall MT-LINK*i* Integration Server.	A-41087 -00001EN	
Integration Server AdminTool OPERATOR'S MANUAL	This manual describes the operation of FANUC MT-LINK*i* Integration Server AdminTool (system administration software).	A-41088 -00001EN	
Integration Server WebUI OPERATOR'S MANUAL	This manual describes the operation of Integration Server WebUI.	A-41089 -00001EN	

Ed.	Date	Design	Description
17	July 28, 2020	uwano	Revised for MT-LINK*i* version 3.10
16	Jan. 30, 2020	uwano	Revised for MT-LINK*i* version 3.9
15	July 22, 2019	uwano	Revised for MT-LINK*i* version 3.8
14	Mar. 29, 2019	ogawa	Revised
13	Jan. 29, 2019	ogawa	Revised for MT-LINK*i* version 3.7
12	Aug. 01, 2018	ogawa	Revised for MT-LINK*i* version 3.6
11	Jan. 12, 2018	ogawa	Revised for MT-LINK*i* version 3.5

Date Dec.25,2015 | Desig. Ogawa | Check | Apprv. Inoue

Title: FANUC MT-LINK*i* Database Schema
Draw: **A-40950-00001EN**
FANUC CORPORATION Sheet. 1/139

TABLE OF CONTENTS

1 OVERVIEW

This document describes the DB configuration of MT-LINK*i*.
MT-LINK*i* deploys MongoDB for DB management.

MongoDB, developed by MongoDB Inc. (formerly 10gen) in 2007, is an open source database management system classified as a document-oriented NoSQL database.

Features of MongoDB:

- Document-oriented database
- JSON data format
- Schema-less
- Index
- Map/Reduce in query syntax
- Replica Set for database replication*
- Sharding for load balancing*
- GridFS for storing more than 16 MB of files
- Capped Collection feature to impose a limit on database usage

* unsupported by MT-LINK*i*

The following table shows key features of MongoDB and RDB (relational database systems):

RDB	MongoDB
database	database
table	collection
record (row)	document
column	field

Following are descriptions of terminology.

Table1. Description of Database Terminology

Term		Description
MT-LINK*i* DB		The database for MT-LINK*i*
Integration DB		The database for Integration Server
PK	Primary key	The parent in a parent-child relationship between documents in databases
FK	Foreign key	A child in a parent-child relationship between documents in databases
AK	Alternate key	Not in use
IE	Inversion entry	Not in use
NN	Not null	Required data as documents
Entity		A collection of objects
Capped Collection		Restricting usage of disk storage by specifying the maximum size of a collection. When storing data to exceed the specified size, the collection overwrites the oldest data.

Note that the "o" marks in the following entity tables denote that the items (PK, FK, AK, and NN) are valid.

Table2. Description of Data Terminology

Term		Description
L0	Machine	Machine
L1	Equipment	Equipment
Tag	Group	Name for grouped equipment
L1_Pool	Equipment status change point history	Retrieving and storing the logs of the status changes per signal of machines (condition of multiple assign status) that compose the equipment
L1Signal_Pool	Equipment signal change point history	Retrieving and storing the logs of the status changes per signal of machines (simple and multiple assign statuses, excluding condition of multiple assign status) that compose the equipment
updatedate	Date and time	The time value is set in the GMT offset for the time zone specified on Server PC.

Signals from devices (CNC) are assigned to statuses of unit called "equipment" and managed by the unit.
The statuses of equipment can be monitored through Web UI screens.
When consisting of more than one device (CNC), equipment can manage the multiple devices collectively.
A device (CNC) signal is assigned to a status or more in a unit or units of "equipment".

- Data is stored in the database on the basis of "equipment".
- Note that device (CNC) signals unassigned to any equipment should not be stored in the database.

1.1 VERSION OF SOFTWARE

The software used to prepare this document is listed in the following table.

Table3. List of Software

Software	Version	Overview
MongoDB	Version 3.6.17	Database with WiredTiger engine and "zlib" compression library

Shown below are the versions of MongoDB and MongoDB driver used in MT-LINK*i* / Integration Server.

- C#Driver: Used in .NET Framework applications such as AdminTool and Receiver.
- mongoose: Used in Web display.

Table4. Versions of MongoDB and MongoDB drivers used in MT-LINK*i* / Integration Server

#	Version of MT-LINK*i*	Version of Integration Server	Version of MongoDB	Version of C#Driver	Version of mongoose	Version of .NET Framework
1	Before 2.1.0	-	3.0.6	2.0.1.27	3.8.31	4.5.2
2	2.1.0	1.0.0	3.2.3	2.2.3.3	3.8.31	4.5.2
3	2.2.0	1.0.0	3.2.3	2.2.3.3	3.8.31	4.5.2
4	3.0.0	1.1.0	3.2.3	2.2.3.3	3.8.31	4.5.2
5	3.1.0	1.2.0	3.2.3	2.3.0.157	4.6.3	4.6.1
6	3.2.0	1.3.0	3.2.3	2.3.0.157	4.6.3	4.6.1
7	3.3.0	1.4.0	3.2.3	2.3.0.157	4.6.3	4.6.1
8	3.4.0	1.5.0	3.2.3	2.3.0.157	4.6.3	4.6.1
9	3.5.0	1.6.0	3.4.9	2.4.4	4.10.8	4.6.1
10	3.6.0	1.7.0	3.4.13	2.5.0	4.13.11	4.6.1
11	3.7.0	1.8.0	3.6.7	2.7.0	5.2.8	4.7.2
12	3.8.0	1.9.0	3.6.11	2.7.3	5.4.19	4.7.2
13	3.9.0	1.10.0	3.6.13	2.7.3	5.4.22	4.7.2
14	3.10.0	1.11.0	3.6.17	2.7.3	5.4.22	4.7.2

> **NOTE**
> If the driver is an old version, the system may not operate properly depending on the version of MongoDB.

1.2 VERSIONS OF SYSTEMS

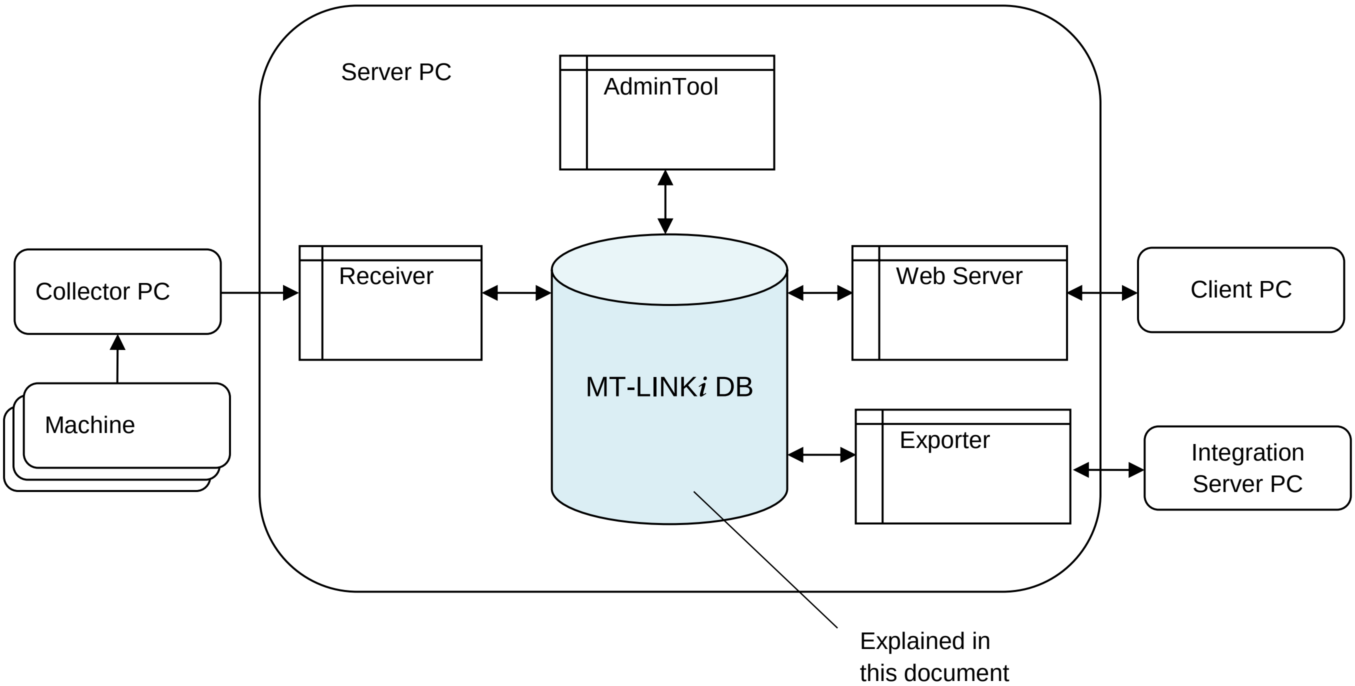
The systems used to prepare this document are listed in the following table.

Table5. List of Systems

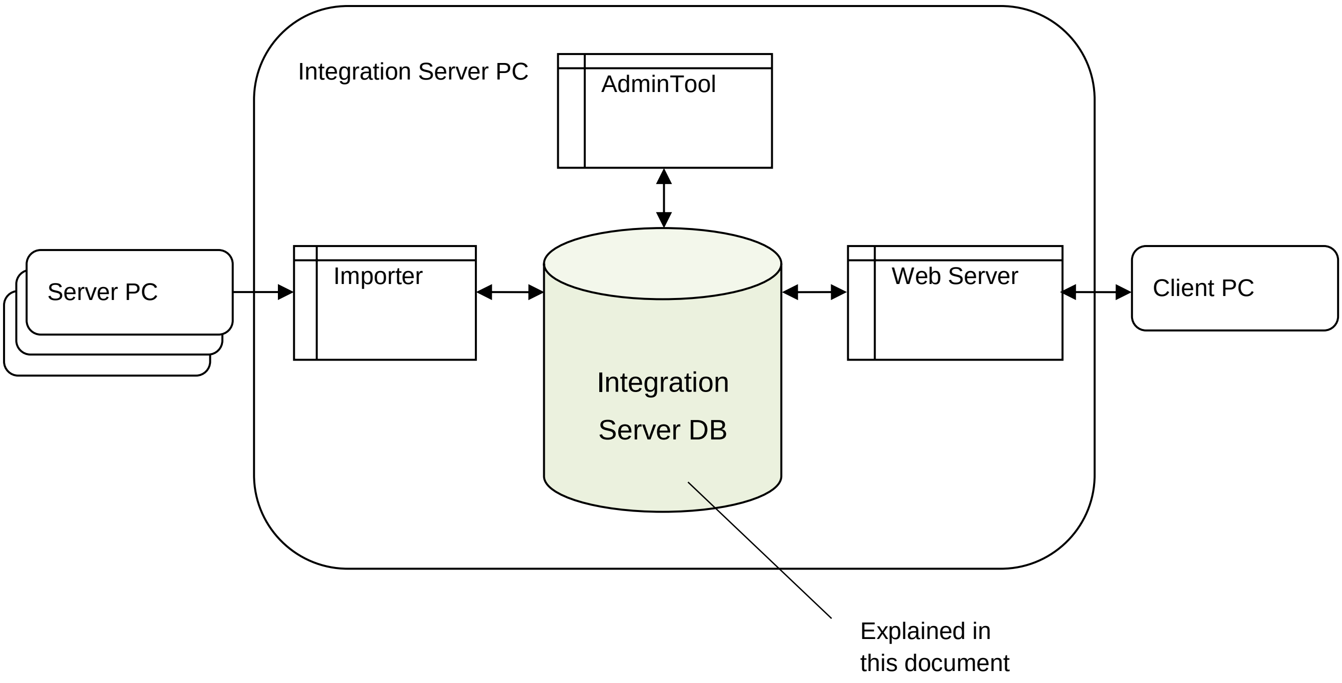
MT-LINK*i* System	Version	Overview
Collector	Version 3.10	Collects information from machines.
AdminTool	Version 3.10	Configures the running environment for MT-LINK*i*.
Receiver	Version 3.10	Stores Collector-collected data in the database.
Web Server	Version 3.10	Displays screens of MT-LINK*i*.
Exporter	Version 1.11	Collects data from MT-LINK*i* database and sends to Importer.
Integration Server System	**Version**	**Overview**
AdminTool	Version 1.11	Configures the running environment for Integration Server.
Importer	Version 1.11	Receives data collected by Exporter and stores in the Integration Server database.
Web Server	Version 1.11	Displays Integration Server WebUI screens.

2 COLLECTION OVERVIEW

2.1 SYSTEM CONFIGURATION



- Collector PC: Collects data from machines and sends the data to Receiver.
- AdminTool: Builds product environment for running MT-LINK*i* and sets up machines and equipment.
- Receiver: Receives the data from Collector PC via network and stores them in the database (DB).
  (Minimum interval: Receiving and storing occur at 500ms or a longer interval.)
- Web Server: To run Web Server, specify the server URL and open WebUI screens.
- Exporter: Receives data from the MT-LINK*i* Database Server and sends data to Integration Server.



- AdminTool: Builds operating environment for running Integration Server.
- Importer: Receives data from Collector PC via network and stores them in the Integration Server database (DB).
- Web Server: To run Web Server, specify the server URL and open WebUI screens.

2.2 LIST OF ENTITIES FOR MT-LINK*i* DB

Table6. List of Entities for MT-LINK*i* DB

#	Entity Name	Physical Name	Definition
1	User Setting	User_Setting	Manages information and languages required for signing in, and assigns roles to users.
2	Authority Setting	Auth_Setting	Assigns authorities for functions, layouts and groups to each role.
3	Layout Setting	Layout_Setting	Manages overview layout information.
4	System Setting	System_Setting	Manages system settings.
5	Collector Setting	Collector_Setting	Manages connection and registration information of the Collector.
6	Machine Setting	L0_Setting	Manages protocol, signal and alarm settings.
7	Equipment Setting	L1_Setting	Manages equipment icons and status assignment.
8	Group Setting	Tag_Setting	Manages equipment units and assign of various statuses.
9	Equipment Status Change Point History	L1_Pool	Records the value of condition received. Also, preserves the status when change occurs on the signal. Registers the time when the change ended and the elapsed time.
10	Equipment Status Change Point History (active)	L1_Pool_Opened	Records the value of condition received. Registers the time when the change started.
11	Equipment Signal Change Point History	L1Signal_Pool	Preserves the start time, end time, and the elapsed time when the change occurred for the equipment name, simple/multiple assign status.
12	Equipment Signal Change Point History (active)	L1Signal_Pool_Active	Preserves the start time when the change occurred for the equipment name, simple/multiple assign status.
13	Group Signal Change Point History	Tag_Pool	Preserves the start time, end time, and elapsed time for the group name and status name (In/Out) when the value started to change.
14	Group Signal Change Point History (active)	Tag_Pool_Active	Preserves the start time for the group name and status name (In/Out) when the value started to change.
15	Alarm History	Alarm_History	Manages alarm history information. Keeps the period of occurrence, time and a variety of information. End Time and Time of Occurrence do not exist during alarm condition.
16	Program History	Program_History	Manages program history information. Keeps the period of execution, the execution time and program names. End Time and Execution Time do not exist during program execution. Running program name does not exist when execution is completed.
17	Macro Variable History	MacroVariableHistory	Manages macro variable history information specified to be obtained by the AdminTool machine setting.
18	Production Plans History	ProductPlan_History	Preserves registered production plans on the basis of equipment name, product name, and production period. Also preserves accumulated production plans.
19	Production Results History	ProductResult_History	Preserves histories of production results on the basis of equipment name, product name, product serial number, and production period. Also preserves added values as accumulated product results. Calculates production result from production result (accumulated).
20	Production Results History (active)	ProductResult_History_Active	Preserves histories of production results on the basis of equipment name, product name, product serial number, and production period. Preserves the start time when the value of production result changed.

#	Entity Name	Physical Name	Definition
21	Work Comments	WorkContents_History	Preserves histories of work comments on the basis of equipment name and period.
22	Operator History	Operator_History	Preserves histories of operator ID on the basis of equipment name and period. Preserves the start time, end time, and period when the operator ID changed.
23	Operator History (active)	Operator_History_Active	Preserves histories of operator ID on the basis of equipment name and period. Preserves the start time when the operator ID changed.
24	Variable History	Variable_History_CNC	Manages variable history information specified for retrieval in AdminTool machine setting.
25	SERVO VIEWER Linkage GridFs File	svFiles.files	Manages the files within GridFS of the files used by SERVO VIEWER linkage.
26	SERVO VIEWER Linkage GridFs Chunk	svFiles.chunks	Manages the chunk of the files expressed by GridFS for the files used by SERVO VIEWER linkage.
27	System Diagnosis	SystemDiagnosis_PC	Manages Receiver/Collector information of system diagnosis.
28	System Diagnosis (Collector)	SystemDiagnosis_Machine	Manages machine information of system diagnosis (Collector).
29	Asset Management Setting	L0Asset_Setting	Manages asset management setting of the machines.
30	Asset Management Online Information	Asset_OnlineInfo	Manages asset management information collected by the Collector and acquired by FOCAS function displayed on the Web screen.
31	Asset Management Online Information Web Name	Asset_OnlineInfoWebName	Manages the Web name of the asset management items displayed on the Web screen.
32	Asset Management Other Systems Linkage	Asset_System_Cooperation	Manages asset management information collected by the Collector and used in linkage with other systems.
33	Servo Axis Configuration Information	AxisConfiguration	Holds servo axis configuration information for machine.
34	LDAP Server Setting	LdapServer_Setting	Holds LDAP Server connection and registration information.

NOTE

The following entities are not used in Version 3.0 or later:
102 Equipment History (L1_History), 111 Group History (Tag_History),
110 Equipment Current Data (L1Live), and 111 Group Current Data (Tag_Live)

2.2.1 User Setting

Table7. User Setting

Entity Name	**Logical Name**	User Setting
	Physical Name	User_Setting
Definition	Manages information required for signing in and the languages, and assigns roles to users. Setting values for the AdminTool [Add/Edit User] dialog box.	

Table8. Definition of User Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	User ID	userid			O	String		1.0	R/W			
2	Username	username	O		O	String		1.0	R/W		R	
3	Password	password			O	String	Encrypt and save	1.0	R/W		R	
4	Language	lang			O	String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	1.0	R/W		R	
5	Role Name	rolenames		O		Array	Assigned to users to control their authorities. Array of role names (String)	1.0	R/W			
6	Calendar Period	calendarSpan		O	O	Array	Used to control period for calendar display. (Array) Refer to "**2.2.1.1 Calendar Period**"	3.0	R/W		R	
7	Web Sign In	webLogin			O	Object	Used to control Web sign in. Refer to "**2.2.1.2 Web Sign In**"	3.4	R/W		R	
8	Authentication Method	AuthMethod			O	Int32	Used to indicate authentication method at sign-in 0: By MT-LINK*i* 1: By LDAP server	3.9	R/W		R/W	
9	Authentication Server	AuthServer			O	String	Server name to be used in authentication	3.9	R/W		R/W	

2.2.1.1 Calendar Period

Table9. Calendar Period

Array Name	**Logical Name**	Calendar Period
	Physical Name	calendarSpan
Definition	Used to control the calendar period displayed. Setting values for the AdminTool [Add/Edit User] dialog box.	

Table10. Definition of Calendar Period

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.0	R/W		R	
2	Function ID	FuncID			O	Int32	ID indicating the function name (NOTE 1)	3.0	R/W		R	
3	Initial Display Period	InitSpan			O	Int32	Initial display period of the calendar period Unit: day Value: 0~7 (0: 0.5 day)	3.0	R/W		R	
4	Maximum Period	MaxSpan			O	Int32	Maximum period of the calendar period Unit: day Value: -1 (NOTE 2), 1~7	3.0	R/W		R	

NOTE

1 Following are the ID's indicating the function names.

100001: Overlook Monitoring
100002: Equipment Monitoring
100003: Group Monitoring
100004: Equipment Monitoring - Detail
100005: Signal Monitoring
100006: Alarm Monitoring
100007: Operational Results
100008: Group Results
100009: Production Results
100010: Alarm History
100011: Signal History
100012: Program History
100013: (unused)
100014: (unused)
100015: Macro Variable History
100016: Report Output
100017: File Transfer
100018: System Diagnosis
100019: SERVO VIEWER Linkage
100020: Machine Data Archive Management

2 The value of the function name "100011: Signal History" is fixed at "-1."

2.2.1.2 Web Sign In

Table11. Web Sign In

Object Name	**Logical Name**	Web Sign In
	Physical Name	webLogin
Definition	Used to control Web sign in procedure. Setting values for the AdminTool [Add/Edit User] dialog box.	

Table12. Definition of Web Sign In

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Skip Sign In	SkipLogin			O	Bool	Enable or disable the skip sign in procedure. Default: false (Disabled)	3.4	R/W		R	

2.2.2 Authority Setting

Table13. Authority Setting

Entity Name	**Logical Name**	Authority Setting
	Physical Name	Auth_Setting
Definition	Assigns authorities for groups, layouts, functions, and system linkage to each role. Setting values for the AdminTool [Add/Edit Role of Authority] dialog box.	

Table14. Definition of Authority Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Role Name	rolename	O		O	String		1.0	R/W		R	
2	Group Authority	tag_auth			O	Array	Authority setting by group Array (tag_auth) Refer to "**2.2.2.1 Group Authority Setting**"	1.0	R/W		R	
3	Layout Authority	layout_auth			O	Array	Authority setting by overlook layout Array (layout_auth) Refer to "**2.2.2.2 Layout Authority Setting**"	1.0	R/W		R	
4	Function Authority	func_auth			O	Array	Authority setting by function Array (func_auth) Refer to "**2.2.2.3 Function Authority Setting**"	1.0	R/W		R	
5	System Linkage Authority	othersystemcoopertaion_auth			O	Array	Authority setting by system linkage Array (othersystemcooperation_auth) Refer to "**2.2.2.4 System Linkage Authority Setting**"	3.1	R/W		R	

2.2.2.1 Group Authority Setting

Table15. Group Authority Setting

Array Name	**Logical Name**	Group Authority Setting
	Physical Name	tag_auth
Definition	Assigns authorities for groups. Setting values for the AdminTool [Add/Edit Role of Authority] dialog box.	

Table16. Definition of Group Authority Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Version	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Name	objectname			O	String	Group name	1.0	R/W		R	
2	Authority	auth_level			O	Int32	Authority setting values 0: No authority 1: ReadOnly 2: Read/Write	1.0	R/W		R	
3	Option	option			O	Int32	Fixed at "0"	1.0	R/W			

2.2.2.2 Layout Authority Setting

Table17. Layout Authority Setting

Array Name	**Logical Name**	Layout Authority Setting
	Physical Name	layout_auth
Definition	Assigns authorities for layouts. Setting values for the AdminTool [Add/Edit Role of Authority] dialog box.	

Table18. Definition of Layout Authority Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Version	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Name	objectname			O	String	Layout name	1.0	R/W		R	
2	Authority	auth_level			O	Int32	Authority setting values 0: No authority 1: ReadOnly 2: Read/Write	1.0	R/W		R	
3	Option	option			O	Int32	Fixed at "0"	1.0	R/W			

2.2.2.3 Function Authority Setting

Table19. Function Authority Setting

Array Name	**Logical Name**	Function Authority
	Physical Name	func_auth
Definition	Assigns authorities for functions. Setting values for the AdminTool [Add/Edit Role of Authority] dialog box.	

Table20. Definition of Function Authority Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Version	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Name	objectname			O	String	Function name	1.0	R/W		R	
2	Authority	auth_level			O	Int32	Authority setting values 0: No authority 1: ReadOnly 2: Read/Write	1.0	R/W		R	
3	Option	option			O	Int32	ID indicating function names (NOTE)	1.0	R/W			

NOTE

Following are the ID's indicating the function names.

100001: Overlook Monitoring
100002: Equipment Monitoring
100003: Group Monitoring
100004: Equipment Monitoring - Detail
100005: Signal Monitoring
100006: Alarm Monitoring
100007: Operational Results
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100010: Alarm History
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100012: Program History
100013: (unused)
100014: (unused)
100015: Macro Variable History
100016: Report Output
100017: File Transfer
100018: System Diagnosis
100019: SERVO VIEWER Linkage
100020: Machine Data Archive Managemen

2.2.2.4 System Linkage Authority Setting

Table21. System Linkage Authority Setting

Array Name	**Logical Name**	System Linkage Authority
	Physical Name	othersystemcooperation_auth
Definition	Assigns authorities for linkage with other systems. Setting values for the AdminTool [Add/Edit Role of Authority] dialog box.	

Table22. Definition of System Linkage Authority Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Version	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Name	objectname			O	String	System name	3.1	R/W		R	
2	Authority	auth_level			O	Int32	Authority setting values 0: No authority 1: ReadOnly	3.1	R/W		R	
3	Option	option			O	Int32	Fixed at "1"	3.1	R/W			

2.2.3 Layout Setting

Table23. Layout Setting

Entity Name	Logical Name	Layout Setting
	Physical Name	Layout_Setting
Definition	Manages overview layout information. Setting values for the AdminTool [Layout Setting] window.	

Table24. Definition of Layout Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Layout Name	LayoutName	O		O	String		1.0	R/W		R	
2	Layout Number	LayoutNo			O	Int32	Layout array number	1.0	R/W		R	
3	Layout Coordinate	background			O	Object	Coordinates of background image Refer to "**2.2.3.1 Background Coordinate**"	1.0	R/W		R	
4	Equipment Coordinate	ObjectList			O	Array	Coordinates of equipment image Refer to "**2.2.3.2 Equipment Coordinate**"	1.0	R/W		R	
5	Status Display	ViewStatus			O	Int32	0: Hide 1: Display	1.0	R/W		R	
6	Auto Alignment Display	AutoEntire			O	Int32	0: Hide 1: Display	3.1	R/W		R	

2.2.3.1 Background Coordinate

Table25. Background Coordinate

Object Name	Logical Name	Background Coordinate
	Physical Name	background
Definition	Manages coordinates of the background image of overlook layout.	

Table26. Definition of Background Coordinate

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Image	uri			O	String	Identifier of background image	1.0	R/W		R	
2	X Coordinate	x			O	Int32	Drawing X position of background image (X axis positive direction)	1.0	R/W		R	
3	Y Coordinate	y			O	Int32	Drawing Y position of background image (Y axis positive direction)	1.0	R/W		R	
4	X Coordinate (negative direction)	regx			O	Int32	Drawing X position of background image (X axis negative direction)	1.0	R/W		R	
5	Y Coordinate (negative direction)	regy			O	Int32	Drawing Y position of background image (Y axis negative direction)	1.0	R/W		R	

2.2.3.2 Equipment Coordinate

Table27. Equipment Coordinate

Array Name	**Logical Name**	Equipment Coordinate
	Physical Name	ObjectList
Definition	Manages coordinates of the equipment image of overlook layout. Setting values for the AdminTool [Layout Setting] window.	

Table28. Definition of Equipment Coordinate

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Equipment ID	id			O	String	Equipment name	1.0	R/W		R	
2	Type	type			O	String	Data format Sprite / bitmap	1.0	R/W		R	
3	Image	uri			O	String	Identifier of equipment image	1.0	R/W		R	
4	Frame	frames			O	Array	Array of equipment image [0]: Operate [1]: Disconnect [2]: Alarm [3]: Emergency [4]: Suspend [5]: Stop [6]: Manual [7]: Warmup Each array above consists of arrays [0]~[3] (Int32), indicating the frame position of each equipment image	1.0	R/W		R	
5	Draw Equipment Image	animations			O	Object	Drawing of equipment image Refer to "**2.2.3.2.1 Draw Equipment Image**"	1.0	R/W		R	
6	Initial Status	initialAction			O	String	Initial status of equipment image Default: DISCONNECT	1.0	R/W		R	
7	X Coordinate	x			O	Int32	Drawing position of equipment image (X axis)	1.0	R/W		R	
8	Y Coordinate	y			O	Int32	Drawing position of equipment image (Y axis)	1.0	R/W		R	
9	X Magnification	scaleX			O	Double	Magnification rate of equipment image (X axis direction)	1.0	R/W		R	
10	Y Magnification	scaleY			O	Double	Magnification rate of equipment image (Y axis direction)	1.0	R/W		R	
11	Status Display	ViewStatus			O	Int32	0: Hide 1: Display * Unused	1.0	R/W		R	

2.2.3.2.1 Draw Equipment Image

Table29. Draw Equipment Image

Object Name	Logical Name	Draw Equipment Image
	Physical Name	animations
Definition	Manages the drawing of equipment image for each status.	

Table30. Definition of Draw Equipment Image

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Operate	OPERATE			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
2	Disconnect	DISCONNECT			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
3	Alarm	ALARM			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
4	Emergency	EMERGENCY			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
5	Suspend	SUSPEND			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
6	Stop	STOP			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
7	Manual	MANUAL			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
8	Warmup	WARMUP			O	Object	frames: Frame position array (Int32) speed: Drawing time (Int32)	1.0	R/W		R	
9	Operate (Blinking)	WOPERATE			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
10	Disconnect (Blinking)	WDISCONNECT			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
11	Alarm (Blinking)	WALARM			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
12	Emergency (Blinking)	WEMERGENCY			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
13	Suspend (Blinking)	WSUSPEND			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
14	Stop (Blinking)	WSTOP			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
15	Manual (Blinking)	WMANUAL			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	
16	Warmup (Blinking)	WWARMUP			O	Object	frames: Frame position array (Int32) speed: Drawing time (Double)	1.0	R/W		R	

2.2.4 System Setting

Table31. System Setting

Entity Name	Logical Name	System Setting
	Physical Name	System_Setting
Definition	Manages the system settings. Setting values for the AdminTool [System Setting] window.	

Table32. Definition of System Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Web Server Root Path	webserver_root			O	String	Web access position Index.html/Login.html	1.0	R/W			
3	Receiver Path	receiver_path			O	String	Full path of Receiver executable file	1.0	R/W			
4	Node Path	node_path			O	String	Full path of Node.js executable file	1.0	R/W			
5	Status	status				String	* Unused	1.0	R/W			
6	Operation History File Path	operation_history_ path				String	* Unused	1.0	R/W			
7	Report File Path	report_path			O	String	Default: <MT-LINKi>\Report	1.0	R/W			
8	Operation History User	Operation_history _ userid					* Unused					
9	Operation History Password	Operation_history _ password					* Unused					
10	Operation History Passive Mode	Operation_history _pasvmode					* Unused					
11	System Version	system_version			O	Object	Version of MT-LINK*i* Refer to "**2.2.4.3 Version Information**"	3.9	R/W			
12	DB Name	DBName			O	String	MongoDB Access Name Default: mongodb://localhost/MTLINKi	1.0	R/W			
13	DB Authentication Username	DBUser			O	String	Username used for DB authentication Default: admin	1.0	R/W			
14	DB Authentication Password	DBPassword			O	String	Password used for DB authentication Default: admin	1.0	R/W			
15	Report UNC Username	report_userid			O	String	Username used for UNC path * Stored after encryption	3.3	R/W			
16	Report UNC Password	report_password			O	String	Password used for UNC path * Stored after encryption	3.3	R/W			
17	Report UNC Permission	report_check			O	Bool	Permission for UNC path true: Permitted false: Not permitted (default)	3.3	R/W			
18	Macro Variable History Storing Path	trigger_path			O	String	Default: <MT-LINKi>\TriggerHistory	3.3	R/W			

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
19	Macro Variable History UNC Username	trigger_userid			O	String	Username used for UNC path * Stored after encryption	3.3	R/W			
20	Macro Variable History UNC Password	trigger_password			O	String	Password used for UNC path * Stored after encryption	3.3	R/W			
21	Macro Variable History UNC Permission	trigger_check			O	Bool	Permission for UNC path true: Permitted false: Not permitted (default)	3.3	R/W			
22	Macro Variable History File Output	trigger_output			O	Bool	Whether to output file or not true: Output false: No output (Default)	3.3	R/W			
23	Macro Variable History Storing Path	variable_history_path			O	String	Default: < MT-LINKi >\VariableHistory	3.3	R/W			
24	Variable History UNC Username	variable_history_userid			O	String	Username used for UNC path * Stored after encryption	3.3	R/W			
25	Variable History UNC Password	variable_history_password			O	String	Password used for UNC path * Stored after encryption	3.3	R/W			
26	Variable History UNC Permission	variable_history_uncpath_check			O	Bool	Permission for UNC path true: Permitted false: Not permitted (Default)	3.3	R/W			
27	Variable History File Output	variable_history_output			O	Bool	Whether to output file or not true: Output false: No output (default)	3.3	R/W			
28	System Linkage	OtherSystemCooperationList			O	Array	Refer to "**2.2.4.1 System Linkage**"	3.1	R/W		R	
29	Header/Footer Color	BackColor			O	String	Background color of header/footer Default: #ffd702	3.1	R/W		R	
30	Show Footer	FooterView			O	String	Show footer text true: Display (default) false: Hide	3.1	R/W		R	
31	Operational Results Display Setting	operation_result_setting			O	Array	Refer to "**2.2.4.2 Operational Results Display Setting**"	3.5	R/W		R	
32	Web Server Port	WebServerPort				Int32	The number of the port used by Web Server Default: 3000	3.8	R/W		R	
33	Web Name Language	webnameLang			O	String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	3.8	R/W			
34	Local Language Name	localLangName				String	Example: "Western European"	3.8	R/W			
35	Local Language Language Code	localLangLanguageCode				String	Example: "fr"	3.8	R/W		R	
36	Local Language Code Page	localLangCodePge				Int32	-1 is set when local language resource is unloaded.	3.8	R/W			
37	Local Language Loaded Date	localLangDate				Date		3.8	R/W			

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access AdminTool	Receiver	Web Server	Exporter
38	Local Language Loaded Version	localLangVersion				String	Example: "3.8.0"	3.8	R/W			

> **NOTE**
> <MT-LINKi> indicates the installation folder of MT-LINK*i*.

2.2.4.1 System Linkage

Table33. System Linkage

Array Name	**Logical Name**	System Linkage
	Physical Name	OtherSystemCooperationList
Definition	Manages settings for the linkage with other systems. Manages the URL of other systems linked on the Web screen and the settings for the Web name and icon displayed on the [Home] screen.	

Table34. Definition of System Linkage

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access AdminTool	Receiver	Web Server	Exporter
1	ID of System	Id			O	String	Unique ID for setting system linkage Value: 01~10	3.1	R/W		R	
2	Name of System (English)	WebEnName			O	String	Web name (English) Default: "NewURL"+System ID	3.1	R/W		R	
3	Name of System (Japanese)	WebJpName			O	String	Web name (Japanese) Default: "NewURL"+System ID	3.1	R/W		R	
4	Name of System (Chinese)	WebCnName			O	String	Web name (Chinese) Default: "NewURL"+System ID	3.1	R/W		R	
5	Name of System (Local language)	WebLocalName			O	String	Web name (Local language) Default: "NewURL"+System ID	3.8	R/W		R	
6	URL of System	Url			O	String	URL default value of other system: http://	3.1	R/W		R	
7	Icon Path	IconPath			O	String	Path of the icon	3.1	R/W		R	

2.2.4.2 Operational Results Display Setting

Table35. Operational Results Display Setting

Array Name	**Logical Name**	Operational Results Display Setting
	Physical Name	operational_result_setting
Definition	Manages operational results display setting. Manages the display/hide status and the order of display for the items (product name, work comments, product serial number, operator ID) displayed on the [Results - Operational results] screen.	

Table36. Definition of Operational Results Display Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Display Item	item			O	String	Items displayed ProductName: Product Name WorkContents: Work Comments ProductSerialNumber: Product Serial Number OperatorID: Operator ID	3.5	R/W		R	
2	Display Order	order			O	Int32	Display order of items Value: 0~3	3.5	R/W		R	
3	Display	display			O	Bool	Whether to display item or not true: Display false: Hide	3.5	R/W		R	
4	Format Version	formatversion			O	Int32	Version of function	3.5	R/W			

2.2.4.3 Version Information

Table37. Version Information

Array Name	Logical Name	Version Information
	Physical Name	Version
Definition	Manages version information of the system. This version information uses Version class defined in .NET Framework.	

Table38. Definition of Version Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Major	Major			O	Int32	Major version number	3.9	R/W		R	
2	Minor	Minor			O	Int32	Minor version number	3.9	R/W		R	
3	Build	Build			O	Int32	Hotfix version number	3.9	R/W		R	
4	Revision	Revision			O	Int32	Revision version number * unused for this system	3.9	R/W		R	
5	Major Revision	MajorRevision			O	Int16	Upper 16 bits of Revision * unused for this system	3.9	R		R	
6	Minor Revision	MinorRevision			O	Int16	Lower 16 bits of Revision * unused for this system	3.9	R		R	

2.2.5 Collector Setting

Table39. Collector Setting

Entity Name	Logical Name	Collector Setting
	Physical Name	Collector_Setting
Definition	Manages connection and registration information of the Collector. Setting values for the AdminTool [Machine Setting] window.	

Table40. Definition of Collector Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Collector Number	CollectorIndex			O	Int32		1.0	R/W			
3	Collector Name	CollectorName			O	String	Computer name	1.0	R/W			
4	Collector IP	CollectorIP			O	String	IP address / host name	1.0	R/W			
5	Collector Port Number	CollectorPort			O	Int32	Default: 43321	1.0	R/W			
6	SERVO VIEWER Linkage Enabled	SVWLinkageFlag			O	Bool	Default: false	3.4	R/W			
7	SERVO VIEWER Linkage Port Number	SVWLinkagePort			O	Int32	Default: 43322	3.4	R/W			
8	Collector Service Manager Port Number	ColMgrPort			O	Int32	Default: 43323	3.8	R/W			
9	Receiver Name	ReceiverName			O	String	Computer name	1.0	R/W			
10	Receiver IP	ReceiverIP			O	String	IP address / host name	1.0	R/W			
11	Receiver Port Number	ReceiverPort			O	Int32	Default: 43320	1.0	R/W			

2.2.6 Machine Setting

Table41. Machine Setting

Entity Name	**Logical Name**	Machine Setting
	Physical Name	L0_Setting
Definition	Manages protocol, signal and alarm settings. Setting values for the AdminTool [Machine Setting] window.	

Table42. Definition of Machine Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Collector Array Number	CollectorIndex			O	Int32		1.0	R/W			
3	Device Location	Index			O	Int32	* Unused	1.0	R/W			
4	Machine Name	L0Name			O	String		1.0	R/W		R	
5	Type	MachineType			O	Int32	0: FanucCNC 5: OPCUAServer 6: FanucRobot 7: MTConnectAgent 8: FanucRobodrill	1.0	R/W		R	
6	Control Type	ControlType				Array	ControlType[x][y] array X [0] to [14]: CNC paths (1 to 15) Y [0]: Laser false: Disable true: Enable	3.5	R/W			
7	Connection Monitoring Status	Connect			O	Bool	Whether to monitor or not true: Monitor (default) false: No monitoring	1.0	R/W		R	
8	Sampling Cycle	SamplingCycle			O	Int32	500 ms (Fixed)	1.0	R/W			
9	Max. Number of Paths for CNC	CncMaxPath			O	Int32	1~15	1.0	R/W			
10	Spindle Unit	SpindleUnit			O	Int32	* Unused	1.0	R/W			
11	Protocol Setting	NetworkSetting			O	Object	Refer to "**2.2.6.1 Protocol Setting**"	1.0	R/W			
12	Signal Setting	Signal_Setting			O	Array	Refer to "**2.2.6.2 Signal Setting**"	1.0	R/W		R	
13	Machine Name (English)	L0EnName			O	String	Web Name (English)	1.1	R/W		R	
14	Machine Name (Japanese)	L0JpName			O	String	Web Name (Japanese)	1.1	R/W		R	
15	Machine Name (Chinese)	L0CnName			O	String	Web Name (Chinese)	1.1	R/W		R	
16	Machine Name (Local language)	L0LocalName			O	String	Web Name (Local language)	3.8	R/W		R	
17	Operation History	OperationHistorySchedule					* Unused					
18	Macro Variable	MacroCaptureDefine				Array	Refer to "**2.2.6.3 Macro Variable Configuration**"	1.0	R/W		R	
19	Variable History	TriggerDefine				Array	Refer to "**2.2.6.4 Variable History Configuration**"	3.3	R/W	R		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
20	Node ID of Selected Machine	ChoseMachineNodeID				String	OPCUAServer only	3.0	R/W			
21	Robot Controller Type	RobotController			O	Int32	FanucRobot only	3.1	R/W			
22	MTConnectAgent Device Name	MTConnectDeviceName			O	String	MTConnectAgent only	3.4	R/W			
23	MTConnectAgent Device ID	MTConnectUuid			O	String	MTConnectAgent only	3.4	R/W			
24	MTConnectAgent Device Setting	Condition_Setting				Object	Refer to "**2.2.6.5 Condition Setting**"	3.8	R/W			

2.2.6.1 Protocol Setting

Table43. Protocol Setting

Object Name	**Logical Name**	Protocol Setting
	Physical Name	NetworkSetting
Definition	Manages protocol settings. Setting values for the AdminTool [Machine Setting] window.	

Table44. Definition of Protocol Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	IP Address	IpAddress			O	String	IP address of machine	1.0	R/W			
2	Port	Port			O	Int32	Communication port of machine	1.0	R/W			
3	Time-out Value	TimeOut			O	Int32	Connection timeout period Default: 30 sec	1.0	R/W			
4	Synchronization Interval	TimeSyncCycle			O	Int32	Time synchronization 0: Not synchronized / 12 / 24 (hours)	3.3	R/W			
5	Access Code Setting Check	AccessCodeCheck			O	Bool	true: Enable false: Disable Default: false	3.10	R/W			
6	Access Code	AccessCode				String	Access code for CNC Stored after encrypted	3.10	R/W			
7	Check Data Server Setting	DataServerCheck			O	Bool	true: Enable false: Disable Default: false	3.7	R/W			
8	Check Data Server IP Address	DataServerIpAddressCheck			O	Bool	true: Enable false: Disable Default: false	3.7	R/W			
9	Data Server IP Address	DataServerIpAddress				String	IP address of transfer destination FTP server	3.7	R/W		R	
10	Data Server Username	DataServerUserName				String		3.7	R/W		R	
11	Data Server Password	DataServerPassword				String	Store after encryption	3.7	R/W		R	
12	Data Server Port Number	DataServerPort			O	Int32	Communication port of transfer destination FTP server Default: 21	3.7	R/W		R	
13	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
14	Protocol Type	datatype			O	Int32	0: TCP/IP (Fixed)	1.0	R/W			

Table45. Definition of Protocol Setting (in the case of OPCUAServer)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Endpoint	EndpointUrl			O	String	Endpoint URL	3.0	R/W			
2	Protocol	Protocol			O	String	Fixed to "opc.tcp"	3.0	R/W			
3	Security Mode	SecurityMode			O	String	"None" "SignAndEncrypt" "Sign"	3.0	R/W			
4	Security Policy	SecurityPolicy			O	String	"None" "Basic128Rsa15" "Basic256"	3.0	R/W			
5	Message Encoding	MessageEncoding			O	String	Fixed to "Binary"	3.0	R/W			
6	User ID	UserIdentity			O	String	"Anonymous" "UserName"	3.0	R/W			
7	User	UserName			O	String	"" (blank), when user ID is "Anonymous" User name (encrypted character string) of the connected OPCUA Server, when user ID is "UserName"	3.0	R/W			
8	Password	Password			O	String	"" (blank), when user ID is "Anonymous" Password (encrypted character string) of the connected OPCUA Server, when user ID is "UserName"	3.0	R/W			
9	Format Version	FormatVersion			O	Int32	Version of function	3.0	R/W			
10	Protocol Type	datatype			O	Int32	2000: OPC	3.0	R/W			

2.2.6.2 Signal Setting

Table46. Signal Setting

Array Name	**Logical Name**	Signal Setting
	Physical Name	Signal_Setting
Definition	Manages signal settings. Setting values for the AdminTool [Machine Setting] window. Signal setting values of FanucCNC / OPCUAServer / FanucRobot / MTConnectAgent.	

Table47. Definition of Signal Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Signal Name	SignalName			O	String	NOTE	1.0	R/W			
3	Signal Configuration	SignalLabel			O	Object	Refer to "**2.2.6.2.1 Signal Configuration**"	1.0	R/W			
4	Outlier Detection Setting	Filter_Setting				Object	Refer to "**2.2.6.2.2 Outlier Detection Setting**"	1.0	R/W		R	
5	Signal Type	SignalType			O	Int32	Signal data type 0: Common 1: Message	1.0	R/W			
6	Data Type of Signal	Type			O	String	Data type of signal Bool Int32 Double String	1.0	R/W			
7	Message	MessageSetting				Object	* Unused	1.0	R/W			
8	Signal Name (Japanese)	SignalJpName			O	String	Web name (Japanese)	1.1	R/W		R	
9	Signal Name (English)	SignalEnName			O	String	Web name (English)	1.1	R/W		R	
10	Signal Name (Chinese)	SignalCnName			O	String	Web name (Chinese)	1.1	R/W		R	
11	Node ID	NodeId				String	Used for OPC UA, MTConnectAgent	3.4	R/W			
12	Display Name	DisplayName				String	Used for OPC UA	3.9	R/W			
13	Original Signal Type	OriginalType				String	Used for OPC UA	3.9	R/W			
14	Method Setting	MethodSetting				Object	Used for OPC UA Refer to "**2.2.6.2.3 Method Setting**"	3.9	R/W			

NOTE

For details on the obtained signals, refer to the "LIST OF FanucCNC SIGNALS" and "LIST OF FanucRobot SIGNALS" subsections of the "FANUC MT-LINK*i* AdminTool OPERATOR'S MANUAL."

2.2.6.2.1 Signal Configuration

Table48. Signal Configuration

Object Name	**Logical Name**	Signal Configuration
	Physical Name	SignalLabel
Definition	Manages signal configuration settings. Setting values for the AdminTool [Machine Setting] window. Setting values of signals / custom signals of FanucCNC / FanucRobot.	

Table49. Definition of Signal Configuration

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	CNC Signal Label	dcLabelFANUCFormat Version			O	Int32	Fixed at "6"	1.0	R/W			
2	Configuration Name	LabelName			O	String	Signal name, or custom signal type (in case of FanucCNC custom signal) PMC MacroVar PcodeVar	1.0	R/W			
3	Address	Adrs			O	String	Address of signal Used for FanucCNC custom signal (PMC), or Robot custom signal Example: "A," "D" (PMC) "DO," "DI" (Robot)	1.0	R/W			
4	Number	Num			O	Int32	Address numbers/ variable numbers of signals	1.0	R/W			
5	Data Type	DataType			O	Int32	Data type of custom signal (NOTE)	1.0	R/W			
6	Bit	Bit			O	Int32	Bit position	1.0	R/W			
7	CNC Path	CncPath			O	Int32	CNC paths of signal 1~15	1.0	R/W			
8	PMC Path	PmcPath			O	Int32	PMC path 1-9	2.1	R/W			
9	Reading Cycle	ReadCycle			O	Int32	Reading cycle 500 ms 10 s 60 s	2.1	R/W			
10	Reading Time	ReadTime			O	Int32	Fixed at "0"	2.1	W			
11	Custom	isCustom			O	Bool	Signal type setting: true: Custom signal setting false: Basic signal setting	1.0	R/W			
12	Option Signal Type	OptionSignalKind			O	Int32	Signal Type 0: Basic/custom signal 1: L (laser signal) 3: OPC method signal	3.5	R/W			
13	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
14	Device Type	LabelType			O	Int32	0:Fanuc (Fixed)	1.0	R/W			
15	Enable	Enable			O	Bool		1.0	R/W			

NOTE

Following are the data types (DataType).

1. In the case of FanucCNC custom signals (PMC)
   - 0: BIT
   - 1: BYTE
   - 2: WORD
   - 3: DWORD
   - 4: REAL (single precision real type)
   - 5: LREAL (double precision real type)
   - 6: Unused
   - 7: Unsigned BYTE
   - 8: Unsigned WORD
   - 9: Unsigned DWORD

2. In the case of FanucCNC custom signals (PcodeVar)
   - 100: Conversation macro
   - 101: Auxiliary macro
   - 102: Execution macro

3. In the case of Robot custom signal
   - 1000: Bool
   - 1001: Int32

4. In the case of MTConnect
   - 1001: Int32
   - 1002: UInt32
   - 1003: String
   - 1004: Double

5. Other than above
   - -1: None

2.2.6.2.2 Outlier Detection Setting

Table50. Outlier Detection Setting

Object Name	**Logical Name**	Outlier Detection Setting
	Physical Name	Filter_Setting
Definition	Manages outlier detection setting of signals. Setting values for the AdminTool [Outlier Detection Setting] dialog box.	

Table51. Definition of Outlier Detection Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Outlier Detection Format Version	dcOverThreaholdFormatVersion			O	Int32	Function version of outlier detection	1.0	R/W			
2	Error Range	error			O	Array	* Unused	1.0	R/W			
3	Warning Range	warnig			O	Array	* Unused	1.0	R/W			
4	Number of Smoothing Data	windowcount			O	Int32	Number of data for smoothing Setting value: 1~100 1: No smoothing	1.0	R/W			
5	Parameter Setting Value	PluginArgs			O	String	Simple Threshold Checker Parameter setting value (NOTE)	1.0	R/W			
6	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
7	Filter Type	FilterKind			O	Int32	Outlier detection type 10001: Simple Threshold Checker	1.0	R/W			
8	Filter Name	PluginName			O	String	Simple Threshold Checker	1.0	R/W			

NOTE

Parameter setting value (PluginArgs) is an array of the following setting values.

- Upper limit value of warning range
- Enable/disable upper limit value of warning range
- Lower limit value of warning range
- Enable/disable lower limit value of warning range
- Upper limit value of error range
- Enable/disable upper limit value of error range
- Lower limit value of error range
- Enable/disable lower limit value of error range
- Number of smoothing data

Example: 8.00, False, 1.00, False, 12.00, False, 0.00, False, 1

2.2.6.2.3 Method Setting

Table52. Method Setting

Object Name	Logical Name	Method Setting
	Physical Name	MethodSetting
Definition	Manages method setting for OPC UA. Setting values for the AdminTool [Method Signal Setting] window.	

Table53. Definition of Method Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Method Setting Format Version	FormatVersion			O	Int32	Function version of method setting	3.9	R/W			
2	Object ID	ObjectId			O	String	Parent ID of the method	3.9	R/W			
3	Method Name	MethodName			O	String	Name of the method	3.9	R/W			
4	Input Arguments Setting	InputArguments			O	Array	Refer to "**Input Arguments Setting**"	3.9	R/W			
5	Output Arguments Setting	OutputArguments			O	Object	Refer to"**Output Arguments Setting**"	3.9	R/W			

2.2.6.2.4 Input Arguments Setting

Table54. Input Arguments Setting

Object Name	**Logical Name**	Input Arguments Setting
	Physical Name	InputArguments
Definition	Manages input arguments setting for OPC UA method signal. Setting values for the AdminTool [Method Signal Setting] window.	

Table55. Definition of Input Arguments Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Input Arguments Format Version	FormatVersion			O	Int32	Function version of input arguments	3.9	R/W			
2	Name	ArgName			O	String	Name of argument	3.9	R/W			
3	Data Type	DataType			O	String	Data type of argument Type set for OPC UA Server	3.9	R/W			
4	Array Rank	ArrayCnt			O	Int32	-1: Scalar	3.9	R/W			
5	Input Value	Value			O	Object	Value input for the argument	3.9	R/W			

2.2.6.2.5 Output Arguments Setting

Table56. Output Arguments Setting

Object Name	Logical Name	Output Arguments Setting
	Physical Name	OutputArguments
Definition	Manages output arguments setting for OPC UA method signal. Setting values for the AdminTool [Method Signal Setting] window.	

Table57. Definition of Output Arguments Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Output Arguments Format Version	FormatVersion			O	Int32	Function version of output arguments	3.9	R/W			
2	Data Position	Index			O	String	Argument setting Index number	3.9	R/W			
3	Name	ArgName			O	String	Name of argument	3.9	R/W			
4	Data Type	DataType			O	String	Data type of argument Type set for OPC UA Server	3.9	R/W			
5	Array Rank	ArrayCnt			O	Int32	-1: Scalar	3.9	R/W			

2.2.6.3 Macro Variable Configuration

Table58. Macro Variable Configuration

Array Name	**Logical Name**	Macro Variable Configuration
	Physical Name	MacroCaptureDefine
Definition	Manages macro variable configuration settings. Setting values for the AdminTool [Machine Setting] window. Macro Variable Getting Condition Definition setting value of FanucCNC.	

Table59. Definition of Macro Variable Configuration

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Macro ID	ID			O	String		1.0	R/W		R	
3	Trigger Number	TriggerMacro			O	Int32		1.0	R/W		R	
4	CNC Path	CNCPath			O	Int32	Macro path	1.0	R/W		R	
5	Reference Macro Number	CaptureMacro			O	Int32		1.0	R/W		R	
6	Capture Count	CaptureCount			O	Int32	Capture count from reference macro number 1~256	1.0	R/W		R	
7	Variable Name (Japanese)	HeaderListForWebJp			O	Array	Web name (Japanese) * Array for the number of captures (String)	3.1	R/W		R	
8	Variable Name (English)	HeaderListForWebEn			O	Array	Web name (English) * Array for the number of captures (String)	3.1	R/W		R	
9	Variable Name (Chinese)	HeaderListForWebCn			O	Array	Web name (Chinese) * Array for the number of captures (String)	3.1	R/W		R	
10	Variable Name (Local language)	HeaderListForWebLocal			O	Array	Web name (Local language) * Array for the number of captures (String)	3.8	R/W		R	
11	Macro ID (Japanese)	JpID			O	String	Web name (Japanese)	1.1	R/W		R	
12	Macro ID (English)	EnID			O	String	Web name (English)	1.1	R/W		R	
13	Macro ID (Chinese)	CnID			O	String	Web name (Chinese)	1.1	R/W		R	
14	Macro ID (Local language)	LocalID			O	String	Web name (Local language)	3.8	R/W		R	
15	Enable	Enable			O	Bool	true: Enable (Fixed)	1.1	R/W		R	

2.2.6.4 Variable History Configuration

Table60. Variable History Configuration

Array Name	**Logical Name**	Variable History Configuration
	Physical Name	TriggerDefine
Definition	Manages variable history configuration settings. Setting values for the AdminTool [Machine Setting] window. Variable History Retrieval Condition Definition setting value of FanucCNC.	

Table61. Definition of Variable History Configuration

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.3	R/W			
2	Variable History ID	TriggerID			O	String	ID of variable history	3.3	R/W	R		
3	Variable History ID (Japanese)	IDJp			O	String	Web name (Japanese)	3.3	R/W	R		
4	Variable History ID (English)	IDEn			O	String	Web name (English)	3.3	R/W	R		
5	Variable History ID (Chinese)	IDCn			O	String	Web name (Chinese)	3.3	R/W	R		
6	Variable History ID (Local language)	IDLocal			O	String	Web name (Local language)	3.8	R/W	R		
7	Trigger Type	TriggerType			O	Int32	Variable type 1: MacroVar (Fixed)	3.3	R/W			
8	Address	TriggerAddress			O	String	When MacroVar is selected: "" (null)	3.3	R/W			
9	Trigger Variable	TriggerNumber			O	Int32	Number of trigger variable	3.3	R/W			
10	Path	Path			O	Int32	Path of CNC	3.3	R/W			
11	Enable	Enable			O	Bool	true: Enable (Fixed)	3.3	R/W			
12	Retrieved Variable Setting	VariableSetting			O	Array	Refer to "**2.2.6.4.1 Retrieved Variable Setting**"	3.3	R/W			

2.2.6.4.1 Retrieved Variable Setting

Table62. Retrieved Variable Setting

Array Name	**Logical Name**	Retrieved Variable Setting
	Physical Name	VariableSetting
Definition	Manages settings of variables retrieved by variable history. Setting values for the AdminTool [Machine Setting] window. Variable History Retrieval Condition Definition setting value of FanucCNC.	

Table63. Definition of Retrieved Variable Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.3	R/W			
2	Type	Type			O	Int32	Basic variable type 0: PMC 1: MacroVar 2: PcodeVar	3.3	R/W			
3	Address	Address			O	String	Used for PMC only When MacroVar / PcodeVar is selected: "" (null)	3.3	R/W			
4	Reference Number	Number			O	Int32	Reference number when retrieving	3.3	R/W			
5	Bit	Bit			O	Int32	Bit position "-1" when other than PMC BIT	3.3	R/W			
6	Variable Count	VariableCount			O	Int32	Retrieval count from reference number 1~256	3.3	R/W			
7	Data Type	DataType			O	Int32	Data type retrieved < MacroVar > "-1" only < PMC > 0: BIT 1: BYTE 2: WORD 3: DWORD 4: REAL 5: LREAL 7: Unsigned BYTE 8: Unsigned WORD 9: Unsigned DWORD < PcodeVar > 100: Conversation macro 101: Auxiliary macro 102: Execution macro	3.3	R/W			
8	Path	Path			O	Int32	Path of CNC	3.3	R/W			
9	Retrieved Variable Name (Japanese)	VariableJp			O	Array	Web name (Japanese) * Array for the number of captures (String) * If no setting: "" (null)	3.3	R/W	R		
10	Retrieved Variable Name (English)	VariableEn			O	Array	Web name (English) * Array for the number of captures (String) * If no setting: "" (null)	3.3	R/W	R		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
11	Retrieved Variable Name (Chinese)	VariableCn			O	Array	Web name (Chinese) * Array for the number of captures (String) * If no setting: "" (null)	3.3	R/W	R		
12	Retrieved Variable Name (Local language)	VariableLocal			O	Array	Web name (Local language) * Array for the number of captures (String) * If no setting: "" (null)	3.8	R/W	R		

2.2.6.5 Condition Setting

Table64. Condition Setting

Array Name	Logical Name	Condition Setting
	Physical Name	Condition_Setting
Definition	Manages condition setting of MTConnectAgent. Setting values for the AdminTool [Machine Setting] window. * Set for MTConnectAgent only.	

Table65. Definition of Condition Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.8	R/W			
2	Output Properties	Output			O	Array	Array (String)	3.8	R/W			
3	Delimiter	Delimiter			O	String	Single alphanumeric character	3.8	R/W			

2.2.7 Equipment Setting

Table66. Equipment Setting

Entity Name	Logical Name	Equipment Setting
	Physical Name	L1_Setting
Definition	Manages equipment icons and status assignment. Setting values for the AdminTool [Equipment Setting] window. When browsed by group setting, group name is retained.	

Table67. Definition of Equipment Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Default Equipment Name	LoadedInitialName			O	String	Used to check if this entry is updated	1.0	R/W			
3	Equipment Name	L1Name	O		O	String		1.0	R/W		R	R
4	Equipment Name (English)	L1EnName			O	String	Web Name (English)	1.1	R/W		R	R
5	Equipment Name (Japanese)	L1JpName			O	String	Web Name (Japanese)	1.1	R/W		R	R
6	Equipment Name (Chinese)	L1CnName			O	String	Web Name (Chinese)	1.1	R/W		R	R
7	Equipment Name (Local language)	L1LocalName			O	String	Web Name (Local language)	3.8	R/W		R	R
8	Equipment Name List	L1Names			O	Array	* Unused	1.0				
9	Group Name	TagNames			O	Array	Array (String) of group name referred to in Group Setting	1.0	R/W		R	R
10	Assign Status Setting	mapped_status			O	Array	Simple assign status Refer to "**2.2.7.1 Status Setting**"	1.0	R/W		R	
11	Standard Status Setting	standard_status			O	Array	Multiple assign status Refer to "**2.2.7.1 Status Setting**"	1.0	R/W		R	
12	Custom Status Setting	custom_status			O	Array	Additional signal setting of multiple assign status Refer to "**2.2.7.1 Status Setting**"	1.0	R/W		R	
13	Deletion Status Setting	deleted_status			O	Array	* Unused	1.0	R/W			
14	Deletion Flag	deleteflag			O	Int32	* Unused	1.0	R/W			
15	Deletion Date and Time	deletedate			O	String	* Unused	1.0	R/W			
16	Count Out Monitoring Flag	notwatched			O	Bool	true: Count out from monitoring false: Not count out (Default)	1.0	R/W		R	
17	Cycle	CycleTime			O	Int32	* Unused	1.0	R/W			
18	DB Reflection	DBStored			O	Bool	true: Sent to DB (Fixed)	1.0	R/W			
19	Operate Image	ImageConnect			O	String	Image storing folder	1.0	R/W		R	
20	Disconnect Image	ImageDisconnect			O	String	Image storing folder	1.0	R/W		R	
21	Alarm Image	ImageAlarm			O	String	Image storing folder	1.0	R/W		R	
22	Emergency Stop Image	ImageEmergency			O	String	Image storing folder	1.0	R/W		R	
23	Suspend Image	ImageSuspend			O	String	Image storing folder	1.0	R/W		R	
24	Stop Image	ImageStop			O	String	Image storing folder	1.0	R/W		R	
25	Manual Image	ImageManual			O	String	Image storing folder	1.0	R/W		R	

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
26	Warmup Image	ImageWarmup			O	String	Image storing folder	1.0	R/W		R	

2.2.7.1 Status Setting

Table68. Status Setting

Array Name	Logical Name	Status Setting
	Physical Name	mapped_status / standard_status / custom_status
Definition	Manages signal status settings. Setting values for the AdminTool [Equipment Setting] window. Setting values for the simple/multiple assign status.	

Table69. Definition of Status Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Priority View	representflag			O	Bool	true: ON false: OFF (default)	1.0	R/W		R	
3	DB Reflection	DBStored			O	Bool	true: Sent to DB (fixed)	1.0	R/W			
4	Status Name	StatusName			O	String	Status names	1.0	R/W		R	
5	Status Name (English)	StatusEnName			O	String	Web Name (English)	1.1	R/W		R	
6	Status Name (Japanese)	StatusJpName			O	String	Web Name (Japanese)	1.1	R/W		R	
7	Status Name (Chinese)	StatusCnName			O	String	Web Name (Chinese)	1.1	R/W		R	
8	Status Name (Local language)	StatusLocalName			O	String	Web Name (Local language)	3.8	R/W		R	
9	Default Status Name	LoadedInitialStatusName			O	String	Used to check if status name is updated	1.0	R/W			
10	Formula for Signal Definition	Formula			O	String	Signal definition formula for signal setting	1.0	R/W			
11	Machine Name	L0Name			O	String		1.0	R/W			
12	Data Type	Type			O	String	Data type of signal Bool Int32 Double String	1.0	R/W			
13	Deletion Flag	deleteflag			O	Bool	* Unused	1.0	R/W			

2.2.8 Group Setting

Table70. Group Setting

Entity Name	Logical Name	Group Setting
	Physical Name	Tag_Setting
Definition	Manages equipment units composing the group and their In/Out assignments. Setting values for the AdminTool [Group Setting] window.	

Table71. Definition of Group Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Default Group Name	LoadedInitialName			O	String	Used to check if this entry is updated	1.0	R/W			
3	Group Name	TagName	O		O	String		1.0	R/W		R	R
4	Group Name (English)	TagEnName			O	String	Web Name (Japanese)	1.1	R/W		R	R
5	Group Name (Japanese)	TagJpName			O	String	Web Name (English)	1.1	R/W		R	R
6	Group Name (Chinese)	TagCnName			O	String	Web Name (Chinese)	1.1	R/W		R	R
7	Group Name (Local language)	TagLocalName			O	String	Web Name (Local language)	3.8	R/W		R	R
8	NORMAL Image	ImageNormal			O	String	NORMAL Image storing folder	1.0	R/W		R	
9	TROUBLE Image	ImageAlarm1			O	String	TROUBLE Image storing folder	1.0	R/W		R	
10	Equipment Name	L1Names			O	Array	Array of equipment names belonging to the group	1.0	R/W		R	
11	Standard Status Setting	standard_status			O	Array	Refer to "**2.2.8.1 Group Status Setting**"	1.0	R/W		R	
12	Custom Status Setting	custom_status			O	Array	* Unused	1.0	R/W			
13	Cycle	CycleTime			O	Int32	* Unused	1.0	R/W			
14	DB Reflection	DBStored			O	Bool	true: Sent to DB (fixed)	1.0	R/W			

2.2.8.1 Group Status Setting

Table72. Group Status Setting

Array Name	**Logical Name**	Group Status Setting
	Physical Name	standard_status
Definition	Manages status setting of groups. Setting values for the AdminTool [Group Setting] window. Setting values for the status (In/Out) and assigned signals.	

Table73. Definition of Group Status Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W			
2	Priority View	representflag			O	Bool	* Unused	1.0	R/W			
3	DB Reflection	DBStored			O	Bool	true: Sent to DB (fixed)	1.0	R/W			
4	Status Name	StatusName			O	String	Status name (In/Out)	1.0	R/W		R	
5	Status Name (English)	StatusEnName			O	String	* Unused	1.0	R/W			
6	Status Name (Japanese)	StatusJpName			O	String	* Unused	1.0	R/W			
7	Status Name (Chinese)	StatusCnName			O	String	* Unused	1.0	R/W			
8	Status Name (Local language)	StatusLoclalName			O	String	* Unused	1.0	R/W			
9	Default Status Name	LoadedInitialStatusName			O	String	Used to check update of status name	1.0	R/W			
10	Formula for Signal Definition	Formula			O	String	Signal definition formula for assigned signals	1.0	R/W			
11	Machine Name	L0Name			O	String	* Unused	1.0	R/W			
12	Data Type	Type			O	String	Data type of status "Int32"	1.0	R/W			
13	Deletion Flag	deleteflag			O	Bool	* Unused	1.0	R/W			

2.2.9 Equipment Status Change Point History

Table74. Equipment Status Change Point History

Entity Name	**Logical Name**	Equipment Status Change Point History
	Physical Name	L1_Pool
Definition	- Records the value of "condition" received. - When the status recorded in the Equipment Status Change Point History (active) (L1_Pool_Opened) changes, the time when the change ended and the elapsed time are registered and recorded. - Records the value of multiple assign status "condition" set on the AdminTool [Equipment Setting] window.	

Table75. Definition of Equipment Status Change Point History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		1.0		R/W	R	R
2	Start Time	updatedate	O		O	Date		1.0		R/W	R	R
3	End Time	enddate	O			Date		1.0		R/W	R	R
4	Time Span	timespan				Double	Unit: Second	1.0		R/W		
5	Signal Name	signalname				String	Multiple assign status "condition"	3.0		R/W		
6	Signal Value	value				String	Character string indicating the state (NOTE)	3.0		R/W	R	R
7	Filter Value	filter				Object	* Unused	1.0		R/W	R	
8	Type	TypeID				Int32	* Unused	1.0		R/W	R	
9	Determined Value	Judge				Int32	* Unused	1.0		R/W	R	
10	Error Threshold	Error				Object	* Unused	1.0		R/W	R	
11	Warning Threshold	Warning				Object	* Unused	1.0		R/W	R	

NOTE

"null" is recorded when machine is disconnected or Collector is stopped.

2.2.10 Equipment Status Change Point History (active)

Table76. Equipment Status Change Point History (active)

Entity Name	Logical Name	Equipment Status Change Point History (active)
	Physical Name	L1_Pool_Opened
Definition	- Records the value of "condition" received. - When the state changes, registers the time when the change started. When the state changes to another state, the ending time and the elapsed time are registered, and move to Equipment Status Change Point History (L1_Pool). - Records the value of multiple assign status "condition" set on the AdminTool [Equipment Setting] window.	

Table77. Definition of Equipment Status Change Point History (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0		R/W	R	R
2	Start Time	updatedate	O		O	Date		2.0		R/W	R	R
3	End Time	Enddate	O			Date	* Unused	2.0		R/W	R	R
4	Time Span	Timespan				Double	* Unused	2.0		R/W		
5	Signal Name	signalname				String	Multiple assign status "condition"	3.0		R/W		
6	Signal Value	value				String	Character string indicating the state (NOTE)	3.0		R/W	R	R
7	Filter Value	filter				Object	* Unused	3.0		R/W		
8	Type	TypeID				Int32	* Unused	3.0		R/W		
9	Determined Value	Judge				Int32	* Unused	3.0		R/W		
10	Error Threshold	Error				Object	* Unused	3.0		R/W		
11	Warning Threshold	Warning				Object	* Unused	3.0		R/W		

NOTE
"null" is recorded when machine is disconnected or Collector is stopped.

2.2.11 Equipment Signal Change Point History

Table78. Equipment Signal Change Point History

Entity Name	**Logical Name**	Equipment Signal Change Point History
	Physical Name	L1Signal_Pool
Definition		- Preserves the start time when the value of simple / multiple assign status started to change and when the change ended and the elapsed time for the equipment. (NOTE 1) - When the state recorded in the Equipment Signal Change Point History (active) (L1Signal_Pool_Opened) changes, the time when the change ended and the elapsed time are registered and recorded. - Records the value of simple/multiple assign status (other than "condition") set on the AdminTool [Equipment Setting] window.

Table79. Definition of Equipment Signal Change Point History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0		R/W	R	
2	Start Time	updatedate	O		O	Date		2.0		R/W	R	
3	End Time	enddate	O			Date		2.0		R/W	R	
4	Time Span	timespan				Double	Unit: Second	2.0		R/W	R	
5	Signal Name	signalname	O			String		3.0		R/W	R	
6	Signal Value	value				variable (NOTE 2)	Value of signal (NOTE 3, 4)	3.0		R/W	R	
7	Filter Value	filter				Double	Converted signal value (Outlier detection value)	3.0		R/W	R	
8	Type	TypeID				Int32	Judging method 10001: Simple Threshold Checker	3.0		R/W	R	
9	Determined Value	Judge				Int32	Judgement result 0: Normal 1: Warning 2: Error	3.0		R/W	R	
10	Error Threshold	Error				Array	Range of error threshold Array [0]: Upper limit (Double) Array [1]: Lower limit (Double)	3.0		R/W	R	
11	Warning Threshold	Warning				Array	Range of alarm threshold Array [0]: Upper limit (Double) Array [1]: Lower limit (Double)	3.0		R/W	R	

NOTE

1. Multiple assign status "condition" is not stored. "condition" is stored in Equipment Status Change Point History / Equipment Status Change Point History (active).
2. The data type becomes variable depending on the signal retrieved. Variable types: Bool / String / Int32 / Double
3. "null" is recorded when machine is disconnected or Collector is stopped.
4. "NaN" is recorded when " " (blank) is collected for the custom signal (custom macro variable).

2.2.12 Equipment Signal Change Point History (active)

Table80. Equipment Signal Change Point History (active)

Entity Name	**Logical Name**	Equipment Signal Change Point History (active)
	Physical Name	L1Signal_Pool_Active
Definition		- Preserves the start time when the value of simple / multiple assign status started to change for the equipment. - When the value changes to another value, the ending time and the elapsed time are registered, and move to Equipment Signal Change Point History (L1Signal_Pool). - Records the value of simple / multiple assign status (other than "condition") set on the AdminTool [Equipment Setting] window.

Table81. Definition of Equipment Signal Change Point History (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0		R/W	R	
2	Start Time	updatedate	O		O	Date		2.0		R/W	R	
3	End Time	enddate				Date	* Unused	2.0		R/W	R	
4	Time Span	timespan				Double	* Unused	2.0		R/W		
5	Signal Name	signalname	O			String		3.0		R/W	R	
6	Signal Value	value				variable (NOTE 1)	Value of signal (NOTE 2, 3)	3.0		R/W	R	
7	Filter Value	filter				Object	Converted signal value (Outlier detection value)	3.0		R/W	R	
8	Type	TypeID				Int32	Judging method 10001: Simple Threshold Checker	3.0		R/W	R	
9	Determined Value	Judge				Int32	Judgement result 0: Normal 1: Warning 2: Error	3.0		R/W	R	
10	Error Threshold	Error				Array	Range of error threshold Array [0]: Upper limit (Double) Array [1]: Lower limit (Double)	3.0		R/W	R	
11	Warning Threshold	Warning				Array	Range of alarm threshold Array [0]: Upper limit (Double) Array [1]: Lower limit (Double)	3.0		R/W	R	

NOTE

1. The data type becomes variable depending on the signal retrieved.
   Variable types: Bool / String / Int32 / Double
2. "null" is recorded when machine is disconnected or Collector is stopped.
3. "NaN" is recorded when " " (blank) is collected for the custom signal (custom macro variable).

2.2.13 Group Signal Change Point History

Table82. Group Signal Change Point History

Entity Name	**Logical Name**	Group Signal Change Point History
	Physical Name	Tag_Pool
Definition	- Preserves the start time, end time, and elapsed time for the group name and status name (In/Out) when the value started to change. - When the status value recorded in the Group Signal Change Point History (active) (Tag_Pool_Active) changes, the time when the change ended and the elapsed time are registered and recorded. - Records the value of status (In/Out) on the AdminTool [Group Setting] window.	

Table83. Definition of Group Signal Change Point History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Group Name	TagName	O		O	String		2.0		R/W	R	
2	Start Time	updatedate	O		O	Date		2.0		R/W	R	
3	End Time	enddate	O			Date		2.0		R/W	R	
4	Time Span	timespan				Double	Unit: Second	2.0		R/W		
5	Signal Name	signalname	O			String	Status name "In" "Out"	3.0		R/W	R	
6	Value	value				Int32	Status value	3.0		R/W	R	

2.2.14 Group Signal Change Point History (active)

Table84. Group Signal Change Point History (active)

Entity Name	**Logical Name**	Group Signal Change Point History (active)
	Physical Name	Tag_Pool_Active
Definition	- Preserves the start time for the group name and status name (In/Out) when the value started to change. - When the value changes to another value, the ending time and the elapsed time are registered, and move to Group Signal Change Point History (Tag_Pool). - Records the value of status (In/Out) on the AdminTool [Group Setting] window.	

Table85. Definition of Equipment Signal Change Point History (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Group Name	TagName	O		O	String		2.0		R/W	R	
2	Start Time	updatedate	O		O	Date		2.0		R/W	R	
3	End Time	enddate	O			Date	* Unused	2.0				
4	Time Span	timespan				Double	* Unused	2.0				
5	Signal Name	signalname	O			String	Status name "In" "Out"	3.0		R/W	R	
6	Value	value				Int32	Status value	3.0		R/W	R	

2.2.15 Alarm History

Table86. Alarm History

Entity Name	Logical Name	Alarm History
	Physical Name	Alarm_History
Definition		Manages alarm history information. Keeps the period of occurrence, time and a variety of information. End Time and Time of Occurrence do not exist during alarm condition.

Table87. Definition of Alarm History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		1.0		R/W	R	R
2	Machine Name	L0Name	O		O	String		1.0		R/W	R	R
3	Number	number	O		O	String		1.0		R/W	R	R
4	Time of Occurrence	updatedate	O		O	Date		1.0		R/W	R	R
5	Message	message	O		O	String		1.0		R/W	R	R
6	End Time	enddate			O	Date		1.0		R/W	R	R
7	Alarm Level	level				Int32	* Unused	1.0		R/W		
8	Alarm Type	type	O			String		1.0		R/W	R	R
9	Time Span of Occurrence	timespan				Double	Unit: Second	1.0		R/W	R	R

2.2.16 Program History

Table88. Program History

Entity Name	Logical Name	Program History
	Physical Name	Program_History
Definition		Manages program history information. Keeps the period of execution, the execution time and program names. End Time and Execution Time are not set during a program execution. When program is already executed, there is no running program name.

Table89. Definition of Program History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		1.0		R/W	R	
2	Machine Name	L0Name	O		O	String		1.0		R/W	R	
3	Path	path	O		O	String	CNC path	1.0		R/W	R	
4	Start Time	updatedate	O		O	Date		1.0		R/W	R	
5	Main Program Flag	mainprogflg	O		O	Bool	History flag true: Main program false: Running program	1.0		R/W	R	
6	End Time	enddate			O	Date	No value during a program execution	1.0		R/W	R	
7	Main Program	mainprog			O	String		1.0		R/W	R	
8	Running Program	runningprog			O	String	In case of Main Program History, the same value as the value of Main Program	1.0		R/W	R	
9	Execution Time	timespan			O	Double	No value during a program execution Unit: Second	1.0		R/W		

2.2.17 Macro Variable History

Table90. Macro Variable History

Entity Name	**Logical Name**	Macro Variable History
	Physical Name	MacroVariableHistory
Definition	Manages macro variable history information specified to acquire in the AdminTool machine setting. Records the history information of Macro Variable Getting Condition Definition for FanucCNC set in the AdminTool [Machine Setting] window.	

Table91. Definition of Macro Variable History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Machine Name	L0Name	O		O	String		1.0		R/W	R	
2	Name	Name	O		O	String	Macro ID name	1.0		R/W	R	
3	Updated Date	UpdateDate	O		O	Date		1.0		R/W	R	
4	Path	Path				String	CNC path	1.0		R/W	R	
5	Trigger Variable	TriggerVariable				String	Trigger macro variable	1.0		R/W	R	
6	Macro Variable	MacroVariable				Array	Refer to "2.2.17.1Macro Variable"	1.0		R/W	R	

2.2.17.1 Macro Variable

Table92. Macro Variable

Array Name	**Logical Name**	Macro Variable
	Physical Name	MacroVariable
Definition	Manages the macro variable data. Records the value of Macro Variable Getting Condition Definition for FanucCNC set in the AdminTool [Machine Setting] window.	

Table93. Definition of Macro Variable

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Name	Name			O	String	Name of retrieved macro variable	1.0		R/W	R	
2	Value	Value			O	Double	Retrieved value	1.0		R/W	R	

2.2.18 Production Plan History

Table94. Production Plan History

Entity Name	**Logical Name**	Production Plan History
	Physical Name	ProductPlan_History
Definition	Preserves histories of registered production plans on the basis of equipment name, product name, and production period. Records the value (product plan) entered on the AdminTool [Manual Data Setting] window.	

Table95. Definition of Production Plan History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0	R/W		R	R
2	Product Name	productname	O		O	String		2.0	R/W		R	R
3	Start Date	updateDate	O		O	Date		2.0	R/W		R	R
4	End Date	enddate	O		O	Date		2.0	R/W		R	R
5	Production Plan	productplan				Int32		2.0	R/W		R	R

2.2.19 Production Results History

Table96. Production Results History

Entity Name	**Logical Name**	Production Results History
	Physical Name	ProductResult_History
Definition	- Preserves histories of production results on the basis of equipment name, product name, product serial number, and production period. The sum of the production results is preserved as the accumulated value. The value of production results is calculated from corresponding accumulated production results. - When the production result recorded in the Production Results History (active) (ProductResult_History_Active) changes, the date / time when the change ended and the elapsed time are registered and recorded. - Records the value of multiple assign status (ProductResultNumber / ProductName/ ProductSerialNumber) set in the AdminTool [Equipment Setting] window. - Records the value of difference when ProductResultNumber increased.	

Table97. Definition of Production Results History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0		R/W	R	R
2	Product Name	productname	O		O	String		2.0		R/W	R	R
3	Start Date	updateDate	O		O	Date		2.0		R/W	R	R
4	End Date	enddate	O		O	Date		2.0		R/W	R	R
5	Elapsed Time	timespan				Double	Unit: second	2.0		R/W		R
6	Production Results	productresult				Int32	Production result (difference value)	2.0		R/W	R	R
7	Production Results (accumulated)	productresult_accumulate				Int32	Production result (accumulated value)	2.0		R/W	R	
8	Production Results Flag	resultflag				Bool	Fixed to "true"	2.0		R/W		
9	Product Serial Number	productserialnumber			O	String		3.5		R/W	R	R

2.2.20 Production Results History (active)

Table98. Production Results History (active)

Entity Name	Logical Name	Production Results History (active)
	Physical Name	ProductResult_History_Active
Definition		- Preserves the start time when the value of production result changed on the basis of equipment name, product name, product serial number, and production period. - When the value changes to another value, the ending time and the elapsed time are registered, and move to Production Results History (ProductResult_History). - Records the value of multiple assign status (ProductResultNumber / ProductName/ ProductSerialNumber) set in the AdminTool [Equipment Setting] window.

Table99. Definition of Production Results History (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.0		R/W		R
2	Product Name	productname	O		O	String		2.0		R/W		R
3	Start Date	updateDate	O		O	Date		2.0		R/W		R
4	End Date	enddate			O	Date	* Unused	2.0		R/W		
5	Elapsed Time	timespan				Double	* Unused	2.0		R/W		R
6	Production Results	productresult				Int32	Production result (difference value)	2.0		R/W		
7	Production Results (accumulated)	productresult_ accumulate				Int32	Production result (accumulated value)	2.0		R/W		
8	Production Results Flag	resultflag				Bool	Fixed to "true"	2.0		R/W		
9	Product Serial Number	productserialn umber			O	String		3.5		R/W		R

2.2.21 Work Comments

Table100.Work Comments

Entity Name		
Entity Name	Logical Name	Work Comments
	Physical Name	WorkContents_History
Definition	Preserves histories of work comments on the basis of equipment name and production period. Records the value (Work Comments) entered on the AdminTool [Manual Data Setting] window.	

Table101.Definition of Work Comments

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		2.1	R/W		R	R
2	Work Comments	workcontents			O	String		2.1	R/W		R	R
3	Start Date	updateDate	O		O	Date		2.1	R/W		R	R
4	End Date	enddate	O		O	Date		2.1	R/W		R	R

2.2.22 Operator History

Table102.Operator History

Entity Name	**Logical Name**	Operator History
	Physical Name	Operator_History
Definition	- Preserves histories of operator ID on the basis of equipment name and period. Preserves the start time, end time, and the elapsed time when the operator ID changed. - When the operator ID recorded in the Operator History (active) (Operator_History_Active) changes, the time when the change ended and the elapsed time are registered and recorded. - Records the value of multiple assign status (OperatorID) specified on the [Equipment Setting] window of AdminTool.	

Table103.Definition of Operator History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		3.5		R/W	R	R
2	Start Date	updatedate	O		O	Date		3.5		R/W	R	R
3	End Date	enddate	O			Date		3.5		R/W	R	R
4	Elapsed Time	timespan				Double	Unit: second	3.5		R/W		
5	Signal Name	signalname				String	Fixed to "OperatorID"	3.5		R/W		
6	Signal Value	value				String	Value of operator ID	3.5		R/W	R	R
7	Filter Value	filter				Object	* Unused	3.5		R/W		
8	Type	TypeID				Int32	* Unused	3.5		R/W		
9	Result	Judge				Int32	* Unused	3.5		R/W		
10	Error Threshold	Error				Object	* Unused	3.5		R/W		
11	Warning Threshold	Warning				Object	* Unused	3.5		R/W		

2.2.23 Operator History (active)

Table104.Operator History (active)

Entity Name	**Logical Name**	Operator History (active)
	Physical Name	Operator_History_Active
Definition	- Preserves histories of operator ID on the basis of equipment name and period. Preserves the start time when the operator ID changed. - When the operator ID changes to another value, the ending time and the elapsed time are registered, and move to Operator History (Operator_History). - Records the value of multiple assign status (OperatorID) specified on the [Equipment Setting] window of AdminTool.	

Table105.Definition of Operator History (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Equipment Name	L1Name	O		O	String		3.5		R/W	R	R
2	Start Date	updatedate	O		O	Date		3.5		R/W	R	R
3	End Date	enddate	O			Date	* Unused	3.5		R/W	R	R
4	Elapsed Time	timespan				Double	* Unused	3.5		R/W		
5	Signal Name	signalname				String	Fixed to "OperatorID"	3.5		R/W		
6	Signal Value	value				String	Value of operator ID	3.5		R/W	R	R
7	Filter Value	filter				Object	* Unused	3.5		R/W		
8	Type	TypeID				Int32	* Unused	3.5		R/W		
9	Result	Judge				Int32	* Unused	3.5		R/W		
10	Error Threshold	Error				Object	* Unused	3.5		R/W		
11	Warning Threshold	Warning				Object	* Unused	3.5		R/W		

2.2.24 Variable History

Table106.Variable History

Entity Name	Logical Name	Variable History
	Physical Name	Variable_History_CNC
Definition	Manages variable history information specified for retrieval in AdminTool machine setting. Records history information of variable history retrieval condition definition of FanucCNC set in the AdminTool [Machine Setting] window.	

Table107.Definition of Variable History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Machine Name	L0Name	O		O	String	L0 Name	3.3		R/W		
2	Name	ID	O		O	String	Variable history ID	3.3		R/W		
3	Trigger Type	TriggerType			O	String	"MacroVar" only	3.3		R/W		
4	Path	TriggerPath			O	Int32	Path of CNC	3.3		R/W		
5	Address	TriggerAddress			O	String	In case of MacroVar: "" (null)	3.3		R/W		
6	Trigger Number	TriggerNumber			O	Int32	Position of trigger variable	3.3		R/W		
7	Retrieval Date	UpdateDate	O		O	Date	Year/month/date of variable retrieval	3.3		R/W		
8	Retrieved Data	VariableData			O	Array	Refer to "**2.2.24.1 Retrieved Data**"	3.3		R/W		

2.2.24.1 Retrieved Data

Table108.Retrieved Data

Array Name	**Logical Name**	Retrieved Data
	Physical Name	VariableData
Definition	Manages variable history information specified for retrieval in AdminTool machine setting. Records history information of retrieved data for variable history retrieval condition definition of FanucCNC set in the AdminTool [Machine Setting] window.	

Table109.Definition of Retrieved Data

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Retrieved Data Type	Type			O	String	< PMC > PMC < Macro variable > MacroVar < P-CODE variable > Common area (100~89999): PcodeVar Conversation macro (1~33): PcodeCon Auxiliary macro (1~33): PcodeAux Execution macro (1~33): PcodeExe	3.3		R/W		
2	Data Type	DataType				Int32	< PMC > 0: bit 1: byte 2: word 3: double-word 4: single precision real 5: double precision real 7: unsigned byte 8: unsigned word 9: unsigned double-word < Macro variable > "null" (unused) < P-CODE variable > 100: conversation macro 101: auxiliary macro 102: execution macro	3.3		R/W		
3	Address	Address			O	String	< PMC > Address of the PMC signal (example: "D" of "D1000") < Other than PMC > "" (empty)	3.3		R/W		
4	Number	Number			O	Int32	< PMC > Location number of the PMC signal (example: "1000" of "D1000") < Other than PMC > Variable number	3.3		R/W		
5	Bit	Bit				Int32	< PMC, bit data type > Bit position (0 to 7) < Otherwise > "null" (unused)	3.3		R/W		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
6	Path	Path			O	Int32	Path of CNC/PMC	3.3		R/W		
7	Value	Value				Object	Retrieved value < PMC > Bit: value of bool Byte: value of Int32 Word: value of Int32 Double-word: value of Int32 Single precision real: value of double Double precision real: value of double Unsigned byte: value of Int64 Unsigned word: value of Int64 Unsigned double-word: value of Int64 < Other than PMC > Value of double-word	3.3		R/W		

2.2.25 SERVO VIEWER Linkage GridFs File

Table110. SERVO VIEWER Linkage GridFs File

Entity Name	**Logical Name**	SERVO VIEWER Linkage GridFs File
	Physical Name	svFiles.files
Definition	Manages the files within GridFS of the files used by SERVO VIEWER linkage. Files managed are the following five types. - L0 setting file for SERVO VIEWER linkage - Measurement schedule - Measure profile - Waveform header - Waveform binary data	

Table111. Definition of SERVO VIEWER Linkage GridFs File

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Receiver	Application Access: Web Server	Application Access: Exporter
1	File Name	filename			O	String	L0 setting file for SERVO VIEWER linkage: L0_SamplingSetting.json Measurement schedule: Machine name_{GUID}.svwsch Measure profile: Machine name_{GUID}.svwraw Waveform header: Machine name_ yyyyMMddTHHmmssZ.svwhdr Waveform binary data: Machine name_ yyyyMMddTHHmmssZ.svwdat	3.4		W	R/W	
2	Date/Time of Upload	uploadDate			O	Date	Date/time of file upload	3.4		W	R/W	
3	Split Size	chunkSize			O	Int64	Fixed to "261120"	3.4		W	R/W	
4	File Size	length			O	Int64	File size	3.4		W	R/W	
5	md5 Hush Info	md5			O	String	MD5 hush of file returned from filemd5 command	3.4		W	R/W	
6	Metadata	metadata			O	Object	Refer to "**2.2.25.1 SERVO VIEWER Linkage GridFs Metadata**"	3.4		W	R/W	
* Following are used in the case of Measurement schedule, Measure profile, Waveform header, or Waveform binary data												
7	Content Type	contentType			O	String	application/octet-stream (fixed)	3.4		W	R/W	
8	Alias Name	aliases			O	Null	null (fixed)	3.4		W	R/W	

2.2.25.1 SERVO VIEWER Linkage GridFs Metadata

Table112. SERVO VIEWER Linkage GridFs Metadata

Object Name	**Logical Name**	SERVO VIEWER Linkage GridFs Metadata
	Physical Name	metadata
Definition	Additional information about GridFs file used by SERVO VIEWER linkage.	

Table113. Definition of SERVO VIEWER Linkage GridFs Metadata

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.4		W	R/W	
2	Machine Device	L0Name			O	String	Machine name (L0 name)	3.4		W	R/W	
3	Date/Time Measurement	MeasurementDateTime				Date	Date/Time of measurement * Set 0 for msec	3.4		W	R	
4	Name of Measure Profile	SvwrawFile				String	Corresponding measure profile name	3.4		W	R	
5	Array Number	Index				Int32	Array number +1 (1~)	3.4			R/W	
6	Search Keyword	SearchKey				String	Search Keyword (corresponds to the search keyword specified in measure schedule)	3.5		W		
7	Target Collector Name	TargetName				String	Collector name (computer name) which the machine belongs to	3.4		W		
8	Waveform binary data sampling state	SamplingResult				Int32	0: complete data -1: incomplete data	3.8		W	R	

The attributes used for each file are shown below. (O: used, X: not used)

	L0 setting file for SERVO VIEWER linkage	Measurement schedule	Measure profile	Waveform header	Waveform binary data
FormatVersion	O	O	O	O	O
L0Name	X	O	O	O	O
MeasurementDateTime	X	X	X	O	O
SvwrawFile	X	X	X	O	O
Index	X	X	O	X	X
SearchKey	X	X	X	O	O
TargetName	O	X	X	X	X
Sampling Result	X	X	X	O	O

2.2.26 SERVO VIEWER Linkage GridFs Chunk

Table114.SERVO VIEWER Linkage GridFs Chunk

Entity Name	**Logical Name**	SERVO VIEWER Linkage GridFs Chunk
	Physical Name	svFiles.chunks
Definition	Manages the separate chunk of the files expressed by GridFS for the files used by SERVO VIEWER linkage. Files managed are the following five types. - L0 setting file for SERVO VIEWER linkage - Measurement schedule - Measure profile - Waveform header - Waveform binary data	

Table115.Definition of SERVO VIEWER Linkage GridFs Chunk

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	File ID	files_id	O		O	ObjectId	_id of the corresponding file collection entry	3.4		W	R/W	
2	Sequential Number	n	O		O	Int32	Sequence number of chunk (0~)	3.4		W	R/W	
3	Data	data			O	Binary	Payload of chunk (BSON binary type)	3.4		W	R/W	

2.2.27 System Diagnosis

Table116. System Diagnosis

Entity Name	**Logical Name**	System Diagnosis
	Physical Name	SystemDiagnosis_PC
Definition	Manages Receiver/Collector information in the system diagnosis. Information about the [System Diagnosis] screen of the Web screen.	

Table117. Definition of System Diagnosis

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	PC Name	PCName	O		O	String	Name of Server PC / Name of Collector PC	3.0		R/W	R	
2	PC Type	PCType	O		O	String	Receiver / Collector	3.0		R/W	R	
3	CPU Usage Rate	CPU			O	Int32	Usage rate of CPU Unit: % Value: 0~100	3.0		R/W	R	
4	Memory Usage Rate	Memory			O	Int32	Usage rate of memory Unit: % Value: 0~100	3.0		R/W	R	
5	Queue Status	QueueStatus			O	Int32	Queue status 1: Good 2: Warning 3: Alarm	3.0		R/W	R	
6	Cycle Time Status	CycleStatus			O	Int32	Cycle time status (NOTE 1) 0: Disconnected / Unknown 1: Good 2: Warning 3: Alarm	3.0		R/W	R	
7	Disk Status	DiskStatus			O	Int32	Disk status 0: Disconnected / Unknown 1: Good 2: Warning 3: Alarm	3.0		R/W	R	
8	Last Update Time	UpdateTime	O		O	Date		3.0		R/W	R	
9	Cache Usage Rate	CacheUsedRate			O	Int32	Usage rate of cache (NOTE 2) Unit: % Value: 0~100 or null (unmeasured)	3.4		R/W	R	
10	Cache Usage Status	CacheStatus			O	Int32	Cache usage status (NOTE 3) 0: Disconnected / Unknown 1: Good 2: Warning 3: Alarm	3.4		R/W	R	
11	DB Data Update Time	CacheUpdateTime			O	String	Last date data stored to DB (NOTE 2) yyyy/MM/dd HH:mm:ss.fff	3.4		R/W	R	
12	Change Points Received	ChangePoint			O	Int32	Number of change points of the signal data retrieved (NOTE 2) Unit: Number of change points / second	3.5		R/W	R	

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
13	Increase of Change Points	DataINC			O	Int32	Increase of change points from last storing of data (NOTE 2) Unit: Number of increase / second	3.5		R/W	R	
14	DB Size	DBDataSize			O	String	* Unused	3.5				
15	Free Disk Space	DiskFree			O	String	Free disk space (GB)	3.6		R/W	R	
16	Free Disk Space Status	DiskFreeStatus			O	Int32	Status of free disk space 0: Disconnected / Unknown 1: Good 2: Warning 3: Alarm 4: Failed to get Disk space	3.6		R/W	R	

NOTE

1 PC type: Only Collector used. Receiver is always "0."
2 PC type: Only Receiver used. Collector is always "null."
3 PC type: Only Receiver used. Collector is always "0."

2.2.28 System Diagnosis (Collector)

Table118. System Diagnosis (Collector)

Entity Name	**Logical Name**	System Diagnosis (Collector)
	Physical Name	SystemDiagnosis_Machine
Definition	Manages machine information in the system diagnosis (Collector). Information about the [System Diagnosis (Collector)] screen of the Web screen.	

Table119. Definition of System Diagnosis (Collector)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	PC Name	PCName			O	String	Name of Collector PC	3.0		R/W	R	
2	Machine Name	MachineName	O		O	String		3.0		R/W	R	
3	Status	Status			O	Int32	0: Disconnected / Unknown 1: Good 2: Warning 3: Alarm	3.0		R/W	R	
4	Cycle Time	CycleTime			O	Int32	Unit: msec Fixed at "500"	3.0		R/W	R	
5	Actual Value of Cycle Time	ExecuteTime			O	Int32	Unit: msec	3.0		R/W	R	
6	Last Update Time	UpdateTime	O		O	Date		3.0		R/W	R	

2.2.29 Asset Management Setting

Table120. Asset Management Setting

Entity Name	**Logical Name**	Asset Management Setting
	Physical Name	L0Asset_Setting
Definition	Manages asset management setting of the machines. Setting values for the AdminTool [Asset Management] window.	

Table121. Definition of Asset Management Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.6	R/W			
2	Machine Name	L0Name			O	String		3.6	R/W		R	
3	Web Title	WebTitle			O	Array	Array (String) [0]: Japanese [1]: English [2]: Chinese [3]: Local language	3.6	R/W		R	
4	Image File Path	ImageFileName			O	Array	Array (String) (NOTE) [0]: Image 1 [1]: Image 2 [2]: Image 3 [3]: Image 4 [4]: Image 5	3.6	R/W		R	
5	Asset Management Items	AssetItemList			O	Array		3.6	R/W		R	
	1 Format Version	FormatVersion			O	Int32	Version of function	3.6	R/W		R	
	2 Web Name	WebName			O	Array	Array (String) [0]: Japanese [1]: English [2]: Chinese [3]: Local language	3.6	R/W		R	
	3 Value	Value			O	Array	Array (String) [0]: Japanese [1]: English [2]: Chinese [3]: Local language	3.6	R/W		R	

NOTE

The relative path from the AdminTool installation folder.
Following is the default AdminTool installation folder.
C:\FANUC\MT-LINKi\MT-LINKiAdminTool

2.2.30 Asset Management Online Information

Table122. Asset Management Online Information

Entity Name	**Logical Name**	Asset Management Online Information
	Physical Name	Asset_OnlineInfo
Definition	Manages asset management information retrieved by FOCAS function displayed on the Web screen.	

Table123. Definition of Asset Management Online Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Receiver	Application Access: Web Server	Application Access: Exporter
1	Machine Name	L0Name	O	O	O	String	L0 name	3.6		W	R	
2	Acquisition Date	UpdateDate			O	Date	Acquisition year/month/date	3.6		W	R	
3	CNC Information	CncInfo			O	Array	Refer to "2.2.30.1 Acquired Information"	3.6		W	R	
4	Path	Path			O	Array	Array of paths (PathInfo)	3.6		W	R	
	1 Path Information	PathInfo			O	Array	Refer to "2.2.30.1 Acquired Information"	3.6		W	R	
5	Collector Acquisition Date	Collector UpdateDate		O	O	Date	Asset information acquisition year/month/date (date/time of Collector PC)	3.6		R/W		

2.2.30.1 Acquired Information

Table124. Acquired Information

Array Name	**Logical Name**	Acquired Information
	Physical Name	Online_Info
Definition	Manages data acquired by FOCAS function displayed on the Web screen.	

Table125. Definition of Acquired Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Item Name	Name			O	String	Name (NOTE)	3.6		W	R	
2	Value	Value			O	String	Value (NOTE)	3.6		W	R	

NOTE

- CNC information
  - [0]: Cnc_Model (CNC model name)
  - [1]: Series (CNC software series)
  - [2]: Version (CNC software version)
  - [3]: Max_axis (The maximum of controlled axes)
  - [4]: Maxpath_no (CNC path)
  - [5]: Eth_type (Ethernet board communication type)
  - [6]: Eth_device (Data Server storage device)
- Path information
  - [0]: mt_type (Control type)
  - [1]: axes (The number of controlled axes)
  - [2]: spdl (The number of controlled spindles)

2.2.31 Asset Management Online Information Web Name

Table126. Asset Management Online Information Web Name

Entity Name	**Logical Name**	Asset Management Online Information Web Name
	Physical Name	Asset_OnlineInfoWebName
Definition	Manages the Web name of the asset management items displayed on the Web screen.	

Table127. Definition of Asset Management Online Information Web Name

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.6	W		R	
2	CNC Information	CncInfo			O	Array	Refer to "**2.2.31.1 Web Name Information**"	3.6	W		R	
3	Path Information	PathInfo			O	Array	Refer to "**2.2.31.1 Web Name Information**"	3.6	W		R	

2.2.31.1 Web Name Information

Table128. Web Name Information

Array Name	**Logical Name**	Web Name Information
	Physical Name	Online_InfoWebName
Definition	Manages the Web name of the asset management items displayed on the Web screen.	

Table129. Definition of Web Name Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.6		W	R	
2	Asset Management Item Name	AssetName			O	String	(NOTE)	3.6		W	R	
3	Web Name (English)	AssetEnName			O	String	(NOTE)	3.6		W	R	
4	Web Name (Japanese)	AssetJpName			O	String	(NOTE)	3.6		W	R	
5	Web Name (Chinese)	AssetCnName			O	String	(NOTE)	3.6		W	R	
6	Web Name (Local language)	AssetLocalName			O	String		3.8		W	R	

NOTE
- CNC information
  - [0]: Cnc_Model (CNC model name)
  - [1]: Series (CNC software series)
  - [2]: Version (CNC software version)
  - [3]: Max_axis (The maximum of controlled axes)
  - [4]: Maxpath_no (CNC path)
  - [5]: Eth_type (Ethernet board communication type)
  - [6]: Eth_device (Data Server storage device)
- Path information
  - [0]: mt_type (Control type)
  - [1]: axes (The number of controlled axes)
  - [2]: spdl (The number of controlled spindles)

2.2.32 Asset Management Other Systems Linkage

Table130. Asset Management Other Systems Linkage

Entity Name	**Logical Name**	Asset Management Other Systems Linkage
	Physical Name	Asset_System_Cooperation
Definition	Manages asset management information.	

Table131. Definition of Asset Management Other Systems Linkage

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Machine Name	L0Name	O	O	O	String	L0 name	3.6		R/W		
2	FOCAS Function Name	FocasName	O	O	O	String	FOCAS function names (cnc_sysinfo / cnc_rdetherinfo / cnc_rdsyssoft2 / cnc_rdmdlconfig2 / cnc_sysinfo_ex / cnc_rdcncid / cnc_rdsyshard / cnc_rdsyssoft3 / cnc_upload4 / cnc_chkversion / cnc_rdservoid2 / cnc_rdfromservoid2 / cnc_rdspindleid2 / cnc_rdfromspindleid2)	3.6		R/W		
3	Acquisition Date	UpdateDate	O		O	Date	Acquisition year/month/date	3.6		R/W		
4	Model Type	Cnc_type			O	Int32	0: 30*i* Series 1: 16*i* Series 2: 15*i* Series	3.6		W		
5	CNC System Information	cnc_sysinfo				Array	Array (cnc_sysinfo) Refer to "**2.2.32.1 CNC System Information**"	3.6		W		
6	Ethernet Board Information	cnc_rdetherinfo				Object	Refer to "**2.2.32.2 Ethernet Board Information**"	3.6		W		
7	CNC System Software Series/Version (Older Models)	cnc_rdsyssoft2				Array	Array (cnc_rdsyssoft2) Refer to "**2.2.32.3 CNC System Software Series/Version Information (Older Models)**"	3.6		W		
8	CNC Module Configuration Information	cnc_rdmdlconfig2				Array	Array (cnc_rdmdlconfig2) Refer to "**2.2.32.4 CNC Module Configuration Information**"	3.6		W		
9	CNC System Information (Extended)	cnc_sysinfo_ex				Object	Refer to "**2.2.32.5 CNC System Information (Extended)**"	3.6		W		
10	CNC Identification Number	cnc_rdcncid				Object	Refer to "**2.2.32.6 CNC ID Number**"	3.6		W		
11	CNC Hardware Configuration Information	cnc_rdsyshard				Object	Refer to ""**2.2.32.7 CNC Hardware Configuration Information**	3.6		W		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
12	CNC System Software Series/Version (New Models)	cnc_rdsyssoft3				Object	Refer to "**2.2.32.8 CNC System Software Series/Version Information (New Models)**"	3.6		W		
13	NC Data Information	cnc_upload4				Object	Refer to "**2.2.32.9 NC Data Information**"	3.6		W		
14	Support Information of FOCAS Function	cnc_chkversion				Object	Refer to "**2.2.32.10 Support Information of FOCAS Function**"	3.7		W	R	
15	SERVO Axis ID Information	cnc_rdservoid2				Array	Refer to "**2.2.32.11 SERVO Axis ID Information**" This one dimensional array holds the data of all the controlled axes starting with the axis number 1.	3.9		W	R	
16	SERVO Axis ID Information in FROM	cnc_rdfromservoid2				Array	Refer to "**2.2.32.12 SERVO Axis ID Information in FROM**" This one dimensional array holds the data of all the controlled axes starting with the axis number 1.	3.9		W	R	
17	Spindle Axis ID Information	cnc_rdspindleid2				Array	Refer to "**2.2.32.13 Spindle Axis ID Information**" This one dimensional array holds the data of all the controlled axes starting with the axis number 1.	3.9		W	R	
18	Spindle Axis ID Information in FROM	cnc_rdfromspindleid2				Array	Refer to "**2.2.32.14 Spindle Axis ID Information in FROM**" This one dimensional array holds the data of all the controlled axes starting with the axis number 1.	3.9		W	R	
19	Collector Acquisition Date/Time	Collector UpdateDate		O	O	Date	Asset information acquisition year/month/date (Collector PC date/time)	3.6		W		
20	Last Receipt Date/Time	LastDate			O	Date	Last receipt date/time Date/time used for DB deletion and DB separation	3.6		R/W		

2.2.32.1 CNC System Information

Table132. CNC System Information

Array Name	Logical Name	CNC System Information
	Physical Name	cnc_sysinfo
Definition	Manages asset management information acquired by cnc_sysinfo.	

Table133. Definition of CNC System Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	CNC Additional Information	addinfo			O	Int32		3.6		W		
2	Max. Controlled Axes	max_axis			O	Int32		3.6		W		
3	Kind of CNC (ASCII)	cnc_type			O	String		3.6		W		
4	Kind of M/T/TT (ASCII)	mt_type			O	String		3.6		W		
5	Series Number (ASCII)	series			O	String		3.6		W		
6	Version Number (ASCII)	version			O	String		3.6		W		
7	Current Controlled Axes (ASCII)	axes			O	String		3.6		W		

2.2.32.2 Ethernet Board Information

Table134. Ethernet Board Information

Object Name	**Logical Name**	Ethernet Board Information
	Physical Name	cnc_rdetherinfo
Definition	Manages asset management information acquired by cnc_rdetherinfo.	

Table135. Definition of Ethernet Board Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Ethernet Board Communication Type	type			O	Int32		3.6		W		
2	Type of Storage Device for Data Server	device			O	Int32		3.6		W		

2.2.32.3 CNC System Software Series/Version Information (Older Models)

Table136. CNC System Software Series/Version Information (Older Models)

Array Name	**Logical Name**	CNC System Software Series/Version Information (Older Models)
	Physical Name	cnc_rdsyssoft2
Definition	Manages asset management information acquired by cnc_rdsyssoft2.	

Table137. Definition of CNC System Software Series/Version Information (Older Models)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Physical Slot Number	slot_no_p				Array	Array (Int32)	3.6		W		
2	Logical Slot Number	slot_no_l				Array	Array (Int32)	3.6		W		
3	Module ID	module_id				Array	Array (Int32)	3.6		W		
4	Software ID	soft_id				Array	Array (Int32)	3.6		W		
5	Software Series with CPU	soft_series				Array	Array (String)	3.6		W		
6	Software Version with CPU	soft_version				Array	Array (String)	3.6		W		
7	Software Installation Status	soft_inst				Int32		3.6		W		
8	Series of Boot Software	boot_ser				String		3.6		W		
9	Version of Boot Software	boot_ver				String		3.6		W		
10	Series of Servo Software	servo_ser				String		3.6		W		
11	Version of Servo Software	servo_ver				String		3.6		W		
12	Series of PMC Management Software	pmc_ser				String		3.6		W		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
13	Version of PMC Management Software	pmc_ver				String		3.6		W		
14	Series of Ladder Software	ladder_ser				String		3.6		W		
15	Version of Ladder Software	ladder_ver				String		3.6		W		
16	Series of Macro Library	mcrlib_ser				String		3.6		W		
17	Version of Macro Library	mcrlib_ver				String		3.6		W		
18	Series of Macro Application	mcrapl_ser				String		3.6		W		
19	Version of Macro Application	mcrapl_ver				String		3.6		W		
20	Series of Spindle Software (1st)	spl1_ser				String		3.6		W		
21	Version of Spindle Software (1st)	spl1_ver				String		3.6		W		
22	Series of Spindle Software (2nd)	spl2_ser				String		3.6		W		
23	Version of Spindle Software (2nd)	spl2_ver				String		3.6		W		
24	Series of Spindle Software (3rd)	spl3_ser				String		3.6		W		
25	Version of Spindle Software (3rd)	spl3_ver				String		3.6		W		
26	Series of C-executer Library	c_exelib_ser				String		3.6		W		
27	Version of C-executer Library	c_exelib_ver				String		3.6		W		
28	Series of C-exe. Application	c_exeapl_ser				String		3.6		W		
29	Version of C-exe. Application	c_exeapl_ver				String		3.6		W		
30	Series of VGA Software (internal)	int_vga_ser				String		3.6		W		
31	Version of VGA Software (internal)	int_vga_ver				String		3.6		W		
32	Series of VGA Software (external)	out_vga_ser				String		3.6		W		
33	Version of VGA Software (external)	out_vga_ver				String		3.6		W		
34	Series of Power Motion Manager Software	pmm_ser				String		3.6		W		
35	Version of Power Motion Manager Software	pmm_ver				String		3.6		W		
36	Series of PMC Management Software	pmc_mng_ser				String		3.6		W		
37	Version of PMC Management Software	pmc_mng_ver				String		3.6		W		
38	Series of PMC Management Software (internal SH)	pmc_shin_ser				String		3.6		W		
39	Version of PMC Management Software (internal SH)	pmc_shin_ver				String		3.6		W		
40	Series of PMC Management Software (external SH)	pmc_shout_ser				String		3.6		W		

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
41	Version of PMC Management Software (external SH)	pmc_shout_ver				String		3.6		W		
42	Series of PMC Management Software (C language)	pmc_c_ser				String		3.6		W		
43	Version of PMC Management Software (C language)	pmc_c_ver				String		3.6		W		
44	Series of PMC Management Software (edit card)	pmc_edit_ser				String		3.6		W		
45	Version of PMC Management Software (edit card)	pmc_edit_ver				String		3.6		W		
46	Series of Ladder Software	lddr_mng_ser				String		3.6		W		
47	Version of Ladder Software	lddr_mng_ver				String		3.6		W		
48	Series of Ladder Software (C application)	lddr_apl_ser				String		3.6		W		
49	Version of Ladder Software (C application)	lddr_apl_ver				String		3.6		W		
50	Series of Spindle Software (4th)	spl4_ser				String		3.6		W		
51	Version of Spindle Software (4th)	spl4_ver				String		3.6		W		
52	Series of 2nd Macro-executer	mcr2_ser				String		3.6		W		
53	Version of 2nd Macro-executer	mcr2_ver				String		3.6		W		
54	Series of 3rd Macro-executer	mcr3_ser				String		3.6		W		
55	Version of 3rd Macro-executer	mcr3_ver				String		3.6		W		
56	Series of Boot Software of Ethernet Board	eth_boot_ser				String		3.6		W		
57	Version of Boot Software of Ethernet Board	eth_boot_ver				String		3.6		W		
58	(reserved)	reserve				Array	Array (String)	3.6		W		
59	Series of Embedded Ethernet Software	embEthe_ser				String		3.6		W		
60	Version of Embedded Ethernet Software	embEthe_ver				String		3.6		W		
61	(reserved)	reserve2				Array	Array (String)	3.6		W		

2.2.32.4 CNC Module Configuration Information

Table138. CNC Module Configuration Information

Array Name	**Logical Name**	CNC Module Configuration Information
	Physical Name	cnc_rdmdlconfig2
Definition	Manages asset management information acquired by cnc_rdmdlconfig2.	

Table139. Definition of CNC Module Configuration Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Data	data				Array	Array of module ID (Int32)	3.6		W		

2.2.32.5 CNC System Information (Extended)

Table140. CNC System Information (Extended)

Object Name	**Logical Name**	CNC System Information (Extended)
	Physical Name	cnc_sysinfo_ex
Definition	Manages asset management information acquired by cnc_sysinfo_ex.	

Table141. Definition of CNC System Information (Extended)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Maximum Controlled Axes	max_axis				Int32		3.6		W		
2	Maximum Spindle Number	max_spdl				Int32		3.6		W		
3	Maximum Path Number	max_path				Int32		3.6		W		
4	Maximum Machining Group Number	max_mchn				Int32		3.6		W		
5	Controlled Axes Number	ctrl_axis				Int32		3.6		W		
6	Servo Axis Number	ctrl_srvo				Int32		3.6		W		
7	Spindle Number	ctrl_spdl				Int32		3.6		W		
8	Path Number	ctrl_path				Int32		3.6		W		
9	Number of Control Machines	ctrl_mchn				Int32		3.6		W		
10	(reserved)	reserved				Int32	Array (Int32)	3.6		W		
11	Data	ODBSYSEX				Array	Array (ODBSYSEX) Refer to "2.2.32.5.1 Path Information (Extended)"	3.6		W		

2.2.32.5.1 Path Information (Extended)

Table142. Path Information (Extended)

Array Name	**Logical Name**	Path Information (Extended)
	Physical Name	ODBSYSEX
Definition	Manages path information acquired by cnc_sysinfo_ex.	

Table143. Definition of Path Information (Extended)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Kind of System	system				Int32		3.6		W		
2	Kind of System Group	group				Int32		3.6		W		
3	Path Attribute	attrib				Int32		3.6		W		
4	Control Axes per Path	ctrl_axis				Int32		3.6		W		
5	Servo Axis Number per Path	ctrl_srvo				Int32		3.6		W		
6	Spindle Number per Path	ctrl_spdl				Int32		3.6		W		
7	Machine Group Number	mchn_no				Int32		3.6		W		
8	(reserved)	reserved				Int32		3.6		W		

2.2.32.6 CNC ID Number

Table144. CNC ID Number

Object Name	**Logical Name**	CNC ID Number
	Physical Name	cnc_rdcncid
Definition	Manages asset management information acquired by cnc_rdcncid.	

Table145. Definition of CNC ID Number

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	CNC ID Number	cncid				Array	Array (Int64)	3.6		W		

2.2.32.7 CNC Hardware Configuration Information

Table146. CNC Hardware Configuration Information

Object Name	**Logical Name**	CNC Hardware Configuration Information
	Physical Name	cnc_rdsyshard
Definition	Manages asset management information acquired by cnc_rdsyshard.	

Table147. Definition of CNC Hardware Configuration Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Hardware Configuration	ODBSYSH_data				Array	Array (ODBSYSH_data) Refer to "**2.2.32.7.1 Hardware Configuration**"	3.6		W		

2.2.32.7.1 Hardware Configuration

Table148. Hardware Configuration

Array Name	**Logical Name**	Hardware Configuration
	Physical Name	ODBSYSH_data
Definition	Manages the hardware configuration.	

Table149. Definition of Hardware Configuration

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	ID-1 (ID)	id1				Int64		3.6		W		
2	ID-2 (Additional Information)	id2				Int64		3.6		W		
3	Group ID	group_id				Int32		3.6		W		
4	Hardware ID	hard_id				Int32		3.6		W		
5	Hardware Number	hard_num				Int32		3.6		W		
6	Slot Number	slot_no				Int32		3.6		W		
7	Display Format for ID-1	id1_format				Int32		3.6		W		
8	Display Format for ID-2	id2_format				Int32		3.6		W		

2.2.32.8 CNC System Software Series/Version Information (New Models)

Table150. CNC System Software Series/Version Information (New Models)

Object Name	**Logical Name**	CNC System Software Series/Version Information (New Models)
	Physical Name	cnc_rdsyssoft3
Definition	Manages asset management information acquired by cnc_rdsyssoft3.	

Table151. Definition of CNC System Software Series/Version Information (New Models)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	System Soft	ODBSYSS3				Array	Array (ODBSYSS3) Refer to "2.2.32.8.1 System Software"	3.6		W		

2.2.32.8.1 System Software

Table152. System Software

Array Name	**Logical Name**	System Software
	Physical Name	ODBSYSS3
Definition	Manages the system software.	

Table153. Definition of System Software

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Software ID	soft_id				Int32		3.6		W		
2	Software Series	soft_series				String		3.6		W		
3	Software Version	soft_edition				String		3.6		W		

2.2.32.9 NC Data Information

Table154. NC Data Information

Object Name	**Logical Name**	NC Data Information
	Physical Name	cnc_upload4
Definition	Manages asset management information acquired by cnc_upload4.	

Table155. Definition of NC Data Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	NC Data	cnc_data				Array	Array (cnc_data) Refer to "2.2.32.9.1 NC Data"	3.6		W		

2.2.32.9.1 NC Data

Table156. NC Data

Array Name	**Logical Name**	NC Data
	Physical Name	cnc_data
Definition	Manages the NC data.	

Table157. Definition of NC Data

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Type of CNC Data	type				Int32	8: Maintenance information	3.6		W		
2	Data	data				String		3.6		W		

2.2.32.10 Support Information of FOCAS Function

Table158. Support Information of FOCAS Function

Object Name	Logical Name	Support Information of FOCAS Function
	Physical Name	cnc_chkversion
Definition	Manages asset management information acquired by cnc_chkversion.	

Table159. Definition of Support Information of FOCAS Function

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Support Information of FOCAS Function	chkversion				Uint32		3.7		W	R	

2.2.32.11 SERVO Axis ID Information

Table160. SERVO Axis ID Information

Object Name	Logical Name	SERVO Axis ID Information
	Physical Name	cnc_rdservoid2
Definition	Manages servo axis ID information acquired with FOCAS function cnc_rdservoid2. Holds ID information per servo axis. (Unretrieved attributes of controlled axes will be filled with space.)	

Table161. Definition of SERVO Axis ID Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Receiver	Web Server	Exporter
1	SERVO Motor Specification	mt_spc				String		3.9		W		
2	SERVO Motor Serial Number	mt_srn				String		3.9		W		
3	Pulse Coder Specification	plc_spc				String		3.9		W		
4	Pulse Coder Serial Number	plc_srn				String		3.9		W		
5	SV Specification	svm_spc				String		3.9		W		
6	SV Serial Number	svm_srn				String		3.9		W		
7	PS Specification	psm_spc				String		3.9		W		
8	PS Serial Number	psm_srn				String		3.9		W		
9	SV Software Series Edition	svs_see				String		3.9		W		
10	PS Software Series Edition	pss_see				String		3.9		W		
11	Separator 1 Specification	pm1_spc				String		3.9		W		
12	Separator 1 Serial Number	pm1_srn				String		3.9		W		
13	Separator 2 Specification	pm2_spc				String		3.9		W		
14	Separator 2 Serial Number	Pm2_srn				String		3.9		W		

2.2.32.12 SERVO Axis ID Information in FROM

Table162. SERVO Axis ID Information in FROM

Object Name	**Logical Name**	SERVO Axis ID Information in FROM
	Physical Name	cnc_rdfromservoid2
Definition	Manages servo axis ID information in FROM acquired with FOCAS function cnc_rdfromservoid2. Holds ID information in FROM per servo axis. (Unretrieved attributes of controlled axes will be filled with space.)	

Table163. Definition of SERVO Axis ID Information in FROM

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	SERVO Motor Specification	mt_spc				String		3.9		W		
2	SERVO Motor Serial Number	mt_srn				String		3.9		W		
3	Pulse Coder Specification	plc_spc				String		3.9		W		
4	Pulse Coder Serial Number	plc_srn				String		3.9		W		
5	SV Specification	svm_spc				String		3.9		W		
6	SV Serial Number	svm_srn				String		3.9		W		
7	PS Specification	psm_spc				String		3.9		W		
8	PS Serial Number	psm_srn				String		3.9		W		
9	SV Software Series Edition	svs_see				String		3.9		W		
10	PS Software Series Edition	pss_see				String		3.9		W		
11	Separator 1 Specification	pm1_spc				String		3.9		W		
12	Separator 1 Serial Number	pm1_srn				String		3.9		W		
13	Separator 2 Specification	pm2_spc				String		3.9		W		
14	Separator 2 Serial Number	Pm2_srn				String		3.9		W		

2.2.32.13 Spindle Axis ID Information

Table164. Spindle Axis ID Information

Object Name	Logical Name	Spindle Axis ID Information
	Physical Name	cnc_rdspindleid2
Definition	Manages spindle axis ID information acquired with FOCAS function cnc_rdspindleid2. Holds ID information per spindle axis. (Unretrieved attributes of controlled axes will be filled with space.)	

Table165. Definition of Spindle Axis ID Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Main SP Motor Specification	mt_spc				String		3.9		W		
2	Main SP Motor Serial Number	mt_srn				String		3.9		W		
3	Sub SP Motor Specification	sbmt_spc				String		3.9		W		
4	Sub SP Motor Serial Number	sbmt_srn				String		3.9		W		
5	SP Specification	spm_spc				String		3.9		W		
6	SP Serial Number	sPm_srn				String		3.9		W		
7	PS Specification	psm_spc				String		3.9		W		
8	PS Serial Number	psm_srn				String		3.9		W		
9	PS Software Series Edition	pss_see				String		3.9		W		

2.2.32.14 Spindle Axis ID Information in FROM

Table166. Spindle Axis ID Information in FROM

Object Name	**Logical Name**	Spindle Axis ID Information in FROM
	Physical Name	cnc_rdfromspindleid2
Definition	Manages spindle axis ID information in FROM acquired with FOCAS function cnc_rdfromspindleid2. Holds D information per spindle axis. (Unretrieved attributes of controlled axes will be filled with space.)	

Table167. Definition of Spindle Axes ID Information in FROM

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Main SP Motor Specification	mt_spc				String		3.9		W		
2	Main SP Motor Serial Number	mt_srn				String		3.9		W		
3	Sub SP Motor Specification	sbmt_spc				String		3.9		W		
4	Sub SP Motor Serial Number	sbmt_srn				String		3.9		W		
5	SP Specification	spm_spc				String		3.9		W		
6	SP Serial Number	spm_srn				String		3.9		W		
7	PS Specification	psm_spc				String		3.9		W		
8	PS Serial Number	psm_srn				String		3.9		W		
9	PS Software Series Edition	pss_see				String		3.9		W		

2.2.33 Servo Axis Configuration Information

Table168. Servo Axis Configuration Information

Entity Name	**Logical Name**	Servo Axis Configuration Information
	Physical Name	AxisConfiguration
Definition	Holds servo axis configuration information for machine.	

Table169.Definition of Servo Axis Configuration Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.9	R/W	R/W		
2	Machine Name	L0Name	O		O	String	L0 name	3.9		R/W	R	
3	Update Date/Time	UpdateDate			O	Date	The date/time when Collector retrieved from NC and updated the data	3.9		R/W	R	
4	Axis Configuration Information	AxisInfo				Array	Array of valid axes (*1)	3.9		R/W	R	
	1 Format Version	FormatVersion				Int32	Version of function	3.9	R/W	R/W	R	
	2 Path Number	Path				Int32	Path number	3.9		R/W	R	
	3 Relative Axis Number in Path	AxisNo_InPath				Int32	Axis number in a path	3.9		R/W	R	
	4 Servo Axis Number	SvAxisNo				Int32	Axis number specified in NC parameters	3.9		R/W	R	
	5 Measurement Axis	LogicalAxisNo				Int32	Axis number to pass to the waveform measurement function (Focas)	3.9		R/W	R	
	6 Axis Name	AxisName				String	Axis name	3.9		R/W	R	

*1 The number of array entries is the sum of the number of spindles and the number of servo axes set as the value of NC parameter No.1023, which can be 1 to the maximum set in max_axis of cnc_sysinfo.

2.2.34 LDAP Server Setting

Table170. LDAP Server Setting

Entity Name	**Logical Name**	LDAP Server Setting
	Physical Name	LdapServer_Setting
Definition	Holds LDAP Server connection and registration information. Setting values for the AdminTool [LDAP Authentication Setting] window.	

Table171. Definition of LDAP Server Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	LDAP Authentication	EnableLdap			O	Bool	true: Enabled false: Disabled	3.9	R/W		R	
2	Automatic LDAP Registration	EnableEntry			O	Bool	true: Enabled false: Disabled	3.9	R/W		R	
3	Server Name	ServerName	O		O	String	Fixed to "LDAPServer"	3.9	R/W		R	
4	IP Address / Host Name	IpAddress			O	String		3.9	R/W		R	
5	Port Number	Port			O	Int32		3.9	R/W		R	
6	LDAPS	Ldaps			O	Bool	*Unused	3.9	R/W		R	
7	Domain	Domain			O	String		3.9	R/W		R	
8	Account	Account			O	String		3.9	R/W		R	
9	Password	Password			O	String	To be encrypted when saved	3.9	R/W		R	
10	Login ID Attribute	LdapID			O	String		3.9	R/W		R	
11	Email Address Attribute	LdapMail			O	String		3.9	R/W		R	
12	Search Scope	BaseDN				String		3.9	R/W		R	
13	LDAP Filter	LdapFilter				String		3.9	R/W		R	
14	Language	lang			O	String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	3.9	R/W		R	
15	Role Name	rolenames		O	O	Array	Assigned to users to control their authorities. Array of role names (String)	3.9	R/W		R	
16	Calendar Period	calendarSpan		O	O	Array	Used to control period for calendar display. (Array) Refer to "**2.2.34.1 Calendar Period**"	3.9	R/W		R	

2.2.34.1 Calendar Period

Table172. Calendar Period

Array Name	**Logical Name**	Calendar Period
	Physical Name	calendarSpan
Definition	Used to control the calendar period displayed. Setting values for the AdminTool [Add/Edit User] dialog box.	

Table173. Definition of Calendar Period

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Receiver	Application Access: Web Server	Application Access: Exporter
1	Format Version	FormatVersion			O	Int32	Version of function	3.0	R/W		R	
2	Function ID	FuncID			O	Int32	ID indicating the function name (NOTE 1)	3.0	R/W		R	
3	Initial Display Period	InitSpan			O	Int32	Initial display period of the calendar period Unit: day Value: 0~7 (0: 0.5 day)	3.0	R/W		R	
4	Maximum Period	MaxSpan			O	Int32	Maximum period of the calendar period Unit: day Value: -1 (NOTE 2), 1~7	3.0	R/W		R	

NOTE

1 Following are the ID's indicating the function names.
- 100001: Overlook Monitoring
- 100002: Equipment Monitoring
- 100003: Group Monitoring
- 100004: Equipment Monitoring - Detail
- 100005: Signal Monitoring
- 100006: Alarm Monitoring
- 100007: Operational Results
- 100008: Group Results
- 100009: Production Results
- 100010: Alarm History
- 100011: Signal History
- 100012: Program History
- 100013: (unused)
- 100014: (unused)
- 100015: Macro Variable History
- 100016: Report Output
- 100017: File Transfer
- 100018: System Diagnosis
- 100019: SERVO VIEWER Linkage
- 100020: Machine Data Archive Management

2 The value of the function name "100011: Signal History" is fixed at "-1."

2.3 LIST OF ENTITIES FOR INTEGRATION SERVER DB

Table174.List of Entities for Integration Server DB

#	Entity Name	Physical Name	Definition
1	User Setting	user_setting	Manages information and languages required for signing in, and assigns roles to users.
2	Role Setting	role_setting	Assigns authorities to roles for displaying.
3	Facility Master	mt_mstr	Manages information about facilities.
4	Group Master	group_mstr	Manages groups.
5	Equipment Master	equipment_mstr	Manages equipment units.
6	Operational Results	operation_result	Stores status of operational results per equipment.
7	Operational Results (active)	operation_result_opened	Stores status of operational results per equipment. Registers the time when the change started.
8	Operational Results Display Setting	operation_result_setting	Manages operational results display setting.
9	Production Results	productresult_history	Stores production results per equipment.
10	Work Comments History	workcontents_history	Stores work comments per equipment.
11	Operator History	operator_history	Stores operator ID per equipment.
12	Facility ID Master	facilityid_mstr	Manages IDs related to facilities
13	Product Plan Number	productplan_history	Stores production plan for each equipment unit.
14	Alarm History	alarm_history	Manages alarm history information.
15	System Linkage Setting	othersystem_setting	Manages system linkage information.
16	System Linkage Role Setting	role_othersystem	Manages system linkage authority information.
17	System Setting	system_version	Manages system information.
18	LDAP Server Setting	ldap_server_setting	Holds LDAP Server connection and registration information.
19	Calendar Period	user_calendar_span	Manages the calendar period of each function for each user.
20	Facility Layout Arrangement Information Role Setting	role_layout	Manages role setting of facility layout arrangement.
21	Facility Layout Arrangement Setting	facilitylayoutarrangement_setting	Manages facility layout arrangement setting.
22	Facility Synchronization Setting	facilitysyncro_setting	Manages facility synchronization setting.

2.3.1 User Setting

Table175.User Setting

Entity Name	**Logical Name**	User Setting
	Physical Name	user_setting
Definition	Manages information required for signing in, and the languages, and assigns roles to users.	

Table176.Definition of User Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	User ID	userid			O	Int32	Automatic numbering from 1	1.0	R/W		
2	User Name	username			O	String	Set user name	1.0	R/W		R
3	Password	password			O	String	Store after encryption	1.0	R/W		R
4	Language	lang			O	String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	1.0	R/W		R
5	Role ID	roleids			O	Array	Used to control access rights Array (String)	1.0	R/W		
6	Format Version	formatversion				Int32	Version of function	1.1	R/W		
7	Authentication Method	AuthMethod			O	Int32	Used to indicate authentication method at sign-in 0: By MT-LINK*i* 1: By LDAP server	1.10	R/W		R/W
8	Authentication Server	AuthServer			O	String	Server name to be used in authentication	1.10	R/W		R/W

2.3.2 Role Setting

Table177. Role Setting

Entity Name	**Logical Name**	Role Setting
	Physical Name	role_setting
Definition	Assigns authorities to roles for displaying.	

Table178. Definition of Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Role ID	roleid			O	Int32	Automatic numbering from 1	1.0	R/W		R
2	Role Name	rolename			O	String	Set role name	1.0	R/W		
3	Functional Role Setting	func_auth			O	Array	Refer to "**2.3.2.1 Functional Role Setting**"	1.0	R/W		R
4	Format Version	formatversion				Int32	Version of function	1.1	R/W		

2.3.2.1 Functional Role Setting

Table179. Functional Role Setting

Array Name	**Logical Name**	Functional Role Setting
	Physical Name	func_auth
Definition	Assigns display authorities per function.	

Table180. Definition of Functional Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Function Name	functionname			O	String	Specify function name (NOTE 1)	1.0	R/W		R
2	Screen Name	pagename			O	String	Specify screen name (NOTE 2)	1.0	R/W		R
3	Authority	auth_level			O	Int32	0: No authority 1: ReadOnly (default)	1.0	R/W		R

NOTE

1 Following are the function names.

facilitymonitoring: Facility Monitoring
operationresult: Operational Results
productresult: Production Results
alarm: Alarm Analysis

2 Following are the screen names.

facilitymonitoring: Facility Monitoring
bargraph: Operational Results - Time Series
piegraph: Operational Results – Ratio Scale
productresult: Production Results
alarm: Alarm Analysis

2.3.3 Facility Master

Table181.Facility Master

Entity Name	Logical Name	Facility Master
	Physical Name	mt_mstr
Definition	Manages information about facilities.	

Table182.Definition of Facility Master

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	MAC Address	macaddress			O	String	Value of MT-LINK*i* Server PC	1.0	R/W	W	R
2	Facility ID	facilityid	O		O	Int64	Automatic numbering from 1	1.0		W	R
3	Facility Name	facilityname				String	Specify facility name	1.0	R	W	R
4	Facility Name (English)	facilityenname				String	Web name (English)	1.0	R	W	R
5	Facility Name (Japanese)	facilityjpname				String	Web name (Japanese)	1.0	R	W	R
6	Facility Name (Chinese)	facilitycnname				String	Web name (Chinese)	1.0	R	W	R
7	Facility Name (Local language)	facilitylocalname				String	Web name (Local language)	1.9	R	W	R
8	Facility URL	facilityurl			O	String	Specify URL	1.7	R	W	R
9	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	R
10	Deletion Date	deletedate				Date	Default: null	1.0		W	R
11	Version	version				Object	Version of Integration Server Refer to "**Version Information**"	1.0		W	
12	Format Version	formatversion				Int32	Version of function	1.1	R	W	
13	Last Collection Date (NOTE)	lastexportdate				Date		1.7		W	R

NOTE

" Last Collection Date " is set as the date/time when the facility transmitted data to the Integration Server. However, there may be cases when the data transmitted is the data that occurred after this date/time, depending on circumstances. Such data may be set in the following collections.

- Operational Results (active) (operation_result_opened)
- Alarm History (alarm_history)
- Operator History (operator_history)

When extracting the data, the data corresponding to the above can be excluded. If the data is to be excluded, make sure that the data excluded is the data of the above corresponding collections with the date/time (updatedate) later than the "Last Collection Date (lastexportdate)" of "Facility Master."

2.3.3.1 Version Information

Table183. Version Information

Array Name	**Logical Name**	Version Information
	Physical Name	Version
Definition	Manages version information of Facility Master. This version information uses Version class defined in .NET Framework.	

Table184. Definition of Version Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Major	Major			O	Int32	Major version number	1.10	R/W		R	
2	Minor	Minor			O	Int32	Minor version number	1.10	R/W		R	
3	Build	Build			O	Int32	Hotfix version number	1.10	R/W		R	
4	Revision	Revision			O	Int32	Revision version number * unused for this system	1.10	R/W		R	
5	Major Revision	MajorRevision			O	Int16	Upper 16 bits of Revision * unused for this system	1.10	R		R	
6	Minor Revision	MinorRevision			O	Int16	Lower 16 bits of Revision * unused for this system	1.10	R		R	

2.3.4 Group Master

Table185. Group Master

Entity Name	Logical Name	Group Master
	Physical Name	group_mstr
Definition	Manages groups.	

Table186. Definition of Group Master

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid	O		O	Int64	Facility ID for the group	1.0	R	W	R
2	Group Name	groupname	O		O	String	Specify the group name	1.0	R	W	R
3	Group Name (English)	groupenname				String	Web name (English)	1.0	R	W	R
4	Group Name (Japanese)	groupjpname				String	Web name (Japanese)	1.0	R	W	R
5	Group Name (Chinese)	groupcnname				String	Web name (Chinese)	1.0	R	W	R
6	Group Name (Local language)	grouplocalname				String	Web name (Local language)	1.9	R	W	R
7	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	R
8	Deletion Date	deletedate				Date	Default: null	1.0		W	
9	Format Version	formatversion				Int32	Version of function	1.1	R	W	

2.3.5 Equipment Master

Table187. Equipment Master

Entity Name	**Logical Name**	Equipment Master
	Physical Name	equipment_mstr
Definition	Manages equipment units.	

Table188. Definition of Equipment Master

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid	O		O	Int64	Facility ID for the equipment	1.0		W	R
2	Group Name	groupname	O		O	String	Specify the group name for the equipment	1.0		W	R
3	Equipment Name	equipmentname			O	String	Specify the equipment name	1.0		W	R
4	Equipment ID	equipmentid	O		O	String	Automatic numbering from 1	1.0		W	R
5	Equipment Name (English)	equipmentenname				String	Web name (English)	1.0		W	R
6	Equipment Name (Japanese)	equipmentjpname				String	Web name (Japanese)	1.0		W	R
7	Equipment Name (Chinese)	equipmentcnname				String	Web name (Chinese)	1.0		W	R
8	Equipment Name (Local language)	equipmentlocalname				String	Web name (Local language)	1.9		W	R
9	Status Setting	standardstatus			O	Array	Refer to "**2.3.5.1 Status Setting**"	1.0		W	R
10	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	
11	Deletion Date	deletedate				Date	Default: null	1.0		W	
12	Format Version	formatversion				Int32	Version of function	1.1		W	

2.3.5.1 Status Setting

Table189. Status Setting

Array Name	Logical Name	Status Setting
	Physical Name	standardstatus
Definition	Manages status settings.	

Table190. Definition of Status Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Status Name	statusname			O	String		1.0		W	R
2	Status Name (English)	statusenname			O	String	Web name (English)	1.0		W	R
3	Status Name (Japanese)	statusjpname			O	String	Web name (Japanese)	1.0		W	R
4	Status Name (Chinese)	statuscnname			O	String	Web name (Chinese)	1.0		W	R
5	Status Name (Local language)	statuslocalname			O	String	Web name (Local language)	1.9		W	R

2.3.6 Operational Results

Table191. Operational Results

Entity Name	**Logical Name**	Operational Results
	Physical Name	operation_result
Definition	Stores status of operational results per equipment. When the status recorded in the Operational Results (active) (operation_result_opened) changes, the time when the change ended is registered and recorded to that data.	

Table192. Definition of Operational Results

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Importer	Application Access: Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.0		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.0		W	R
3	Equipment ID	equipmentid		O	O	String	Equipment ID	1.0		W	
4	Time	updatedate				Date	Set updatedate of L1Signal_Pool (L1Name = Equipment name of item2)	1.0		W	R
5	Status	status				String	Set value of L1Signal_Pool (L1Name = Equipment name of item2)	1.0		W	R
6	End Date	enddate				Date	Set enddate of L1Signal_Pool (L1Name = Equipment name of item2) Set last collection date/time of facility master, if operational results data is active	1.7		W	R
7	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	
8	Deletion Date	deletedate				Date	Default: null	1.0		W	

2.3.7 Operational Results (active)

Table193. Operational Results (active)

Entity Name	**Logical Name**	Operational Results (active)
	Physical Name	operation_result_opened
Definition	Stores status of operational results per equipment. Registers the time when the change started. When the status changes to another status, the ending time is registered, and moves to Operational Results (operation_result).	

Table194. Definition of Operational Results (active)

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Importer	Application Access: Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.0		W	
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.0		W	R
3	Equipment ID	equipmentid		O	O	String	Equipment ID	1.0		W	
4	Time	updatedate				Date	Set updatedate of L1_Pool_Opened (L1Name = Equipment name of item2)	1.0		W	R
5	Status	status				String	Set value of L1_Pool_Opened (L1Name = Equipment name of item2)	1.0		W	R
6	End Date	enddate				Date	Set enddate of L1_Pool_Opened (L1Name = Equipment name of item2)	1.0		W	R
7	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	
8	Deletion Date	deletedate				Date	Default: null	1.0		W	

2.3.8 Operational Results Display Setting

Table195. Operational Results Display Setting

Entity Name	**Logical Name**	Operational Results Display Setting
	Physical Name	operational_result_setting
Definition	Manages operational results display setting. Manages the display/hide status and the order of display for the items (product name, work comments, product serial number, operator ID) displayed on the [Results - Operational results] screen.	

Table196. Definition of Operational Results Display Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Display Item	item			O	String	Items displayed "Product Name" "Work Comments" "Product Serial Number" "Operator ID"	1.6	W		R
2	Display Order	order			O	Int32	Start from "0" (zero)	1.6	W		R
3	Display	display			O	Bool	true: Display, false: Hide	1.6	W		R
4	Format Version	formatversion			O	Int32	Version of function	1.6	W		

2.3.9 Production Results

Table197. Production Results

Entity Name	**Logical Name**	Production Results
	Physical Name	productresult_history
Definition	Stores production results per equipment.	

Table198. Definition of Production Results

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Importer	Application Access: Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.0		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.0		W	R
3	Equipment ID	equipmentid		O	O	String	Equipment ID	1.0		W	R
4	Product Name	productname			O	String	Set productname of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	R
5	Start Date	updatedate			O	Date	Set updatedate of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	R
6	End Date	enddate			O	Date	Set enddate of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	R
7	Production Results	productresult				Int64	Set productresult of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	R
8	Production Results (accumulated)	productresult_accumulate				Int64	Set productresult_accumulate of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	
9	Results Flag	resultflag			O	Bool	Set resultflag of ProductResult_History (L1Name = Equipment name of item2)	1.0		W	
10	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	
11	Deletion Date	deletedate				Date	Default: null	1.0		W	
12	Product Serial Number	productserialnumber			O	String	Set productserialnumber of ProductResult_History (L1Name = Equipment name of item2)	1.6		W	R

2.3.10 Work Comments History

Table199. Work Comments History

Entity Name	Logical Name	Work Comments History
	Physical Name	workcontents_history
Definition	Stores work comments per equipment.	

Table200. Definition of Work Comments History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.0		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.0		W	R
3	Equipment ID	equipmentid		O	O	String	Specify <Facility ID> _<Equipment name>	1.0		W	
4	Start Date	updatedate			O	Date	Set updatedate of WorkContents_History (L1Name = Equipment name of item2)	1.0		W	R
5	End Date	enddate			O	Date	Set enddate of WorkContents_History (L1Name = Equipment name of item2)	1.0		W	R
6	Work Comments	workcontents				String	Set workcontents of WorkContents_History (L1Name = Equipment name of item2)	1.0		W	R
7	Deletion Flag	deleteflag			O	Bool	Default: false	1.0		W	
8	Deletion Date	deletedate				Date	Default: null	1.0		W	

2.3.11 Operator History

Table201. Operator History

Entity Name	**Logical Name**	Operator History
	Physical Name	operator_history
Definition	Stores operator ID per equipment.	

Table202. Definition of Operator History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access: AdminTool	Application Access: Importer	Application Access: Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.6		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.6		W	R
3	Equipment ID	equipmentid		O	O	String	Specify the equipment ID	1.6		W	
4	Start Date/Time	updatedate			O	Date	Set starting date and time of operational results data	1.6		W	R
5	End Date/Time	enddate			O	Date	Set ending date and time of operational results data	1.6		W	R
6	Operator ID	operatorid				String	Set operator ID of operational results data	1.6		W	R
7	Deletion Flag	deleteflag			O	Bool	Default: false	1.6		W	
8	Deletion Date	deletedate				Date	Default: null	1.6		W	

2.3.12 Facility ID Master

Table203. Facility ID Master

Entity Name	Logical Name	Facility ID Master
	Physical Name	facilityid_mstr
Definition	Manages IDs related to facilities.	

Table204. Definition of Facility ID Master

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid	O		O	Int64	Numbered starting from 1	1.0		R/W	
2	User ID	userid			O	Int64	Numbered starting from 1	1.0	R/W		
3	Role ID	roleid			O	Int64	Numbered starting from 1	1.0	R/W		
4	Other System ID	othersystemid			O	Int64	Numbered starting from 1	1.2	R/W		

2.3.13 Production Plan History

Table205. Production Plan History

Entity Name	Logical Name	Production Plan History
	Physical Name	productplan_history
Definition	Stores production plan for each equipment unit	

Table206. Definition of Production Plan History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.1		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.1		W	R
3	Equipment ID	equipmentid			O	String	Equipment ID	1.1		W	
4	Product Name	productname			O	String	Set productname of ProductPlan_History (Product plan history)	1.1		W	R
5	Start Date	updatedate			O	Date	Set updatedate of ProductPlan_History (Product plan history)	1.1		W	R
6	End Date	enddate			O	Date	Set enddate of ProductPlan_History (Product plan history)	1.1		W	R
7	Production Plan	productplan				Int64	Set productplan of ProductPlan_History (Product plan history)	1.1		W	R
8	Production Plan (accumulated)	productplan_accumulate				Int64	Set productplan_accumulate of ProductPlan_History (Product plan history)	1.1		W	
9	Deletion Flag	deleteflag				Bool	Default: false	1.1		W	
10	Deletion Date	deletedate				Date	Default: null	1.1		W	

2.3.14 Alarm History

Table207.Alarm History

Entity Name	Logical Name	Alarm History
	Physical Name	alarm_history
Definition	Manages alarm history information	

Table208. Definition of Alarm History

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid			O	Int64	Facility ID for the equipment	1.1		W	R
2	Equipment Name	equipmentname			O	String	Specify the equipment name	1.1		W	R
3	Equipment ID	equipmentid		O	O	String	Specify <Facility ID> _<Equipment name>	1.1		W	R
4	Machine Name	machinename			O	String	Set L0Name of Alarm_History (Alarm history)	1.1		W	R
5	Number	number			O	String	Set number of Alarm_History (Alarm history)	1.1		W	R
6	Time of Day	updatedate			O	Date	Set updatedate of Alarm_History (Alarm history)	1.1		W	R
7	Message	message			O	String	Set message of Alarm_History (Alarm history)	1.1		W	R
8	End Time	enddate				Date	Set enddate of Alarm_History (Alarm history)	1.1		W	R
9	Alarm Type	type				String	Set type of Alarm_History (Alarm history)	1.1		W	R
10	Time of Occurrence	timespan				Double	Unit: seconds	1.1		W	R
11	Deletion Flag	deleteflag				Bool	Default: false	1.1		W	
12	Deletion Date	deletedate				Date	Default: null	1.1		W	

2.3.15 System Linkage Setting

Table209. System Linkage Setting

Entity Name	Logical Name	System Linkage Setting
	Physical Name	othersystem_setting
Definition	Manages system linkage setting.	

Table210. Definition of System Linkage Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Name of Other System	pagename			O	String	Start from 1	1.2	W		R
2	Order of Display	displayorder			O	Int32	Start from 1	1.2	W		R
3	URL	url			O	String	URL of other system	1.2	W		R
4	Name of Other System (English)	webenname			O	String	Web name (English)	1.2	W		R
5	Name of Other System (Japanese)	webjpname			O	String	Web name (Japanese)	1.2	W		R
6	Name of Other System (Chinese)	webcnname			O	String	Web name (Chinese)	1.2	W		R
7	Name of Other System (Local language)	weblocalname			O	String	Web name (Local language)	1.9	W		R
8	Icon Path	iconpath			O	String	Path of the icon file	1.2	W		R
9	Format Version	formatversion			O	Int32	Version of function	1.2	W		R

2.3.16 System Linkage Role Setting

Table211. System Linkage Role Setting

Entity Name	**Logical Name**	System Linkage Role Setting
	Physical Name	role_othersystem
Definition	Manages system linkage role information.	

Table212. Definition of System Linkage Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Role ID	roleid			O	Int32	Start from 1	1.2	W		
2	Functional Role	func_auth			O	Array	Refer to "**2.3.16.1 Functional Role Setting**"	1.2	R/W		R
3	Format Version	formatversion			O	Int32	Version of function	1.2	W		

2.3.16.1 Functional Role Setting

Table213. Functional Role Setting

Array Name	**Logical Name**	Functional Role Setting
	Physical Name	func_auth
Definition	Assigns role to display.	

Table214. Definition of Functional Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Function Name	functionname			O	String	"OtherSystem"	1.2	R/W		R
2	Screen Name	pagename			O	String		1.2	R/W		R
3	Authority	auth_level			O	Int32	0: No Authority 1: Read Only (Default)	1.2	R/W		R

2.3.17 System Setting

Table215. System Setting

Entity Name	Logical Name	System Setting
	Physical Name	system_version
Definition	Manages system information.	

Table216. Definition of System Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	System Version	System_version			O	Object	Version of Integration Server Refer to "**2.3.17.1 Version Information**"	1.10	R/W		
2	Format Version	FormatVersion			O	Int32	Version of function	1.0	R/W		
3	Header/Footer Color	BackColor			O	String	Default: #ffd702	1.0	R/W		R
4	Show Footer	FotterView			O	String	Default: true	1.0	R/W		R
5	Web Name Language	webnameLang			O	String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	1.9	R/W		
6	Local Language Name	localLangName				String	Example: "Western European"	1.9	R/W		
7	Local Language Language Code	localLangLanguageCode				String	Example: "fr"	1.9	R/W		R
8	Local Language Code Page	localLangCodePge				Int32	-1 is set when local language resource is unloaded.	1.9	R/W		
9	Local Language Loaded Date	localLangDate				Date		1.9	R/W		
10	Local Language Loaded Version	localLangVersion				String	Example: "1.9.0"	1.9	R/W		

2.3.17.1 Version Information

Table217.Version Information

Array Name	**Logical Name**	Version Information
	Physical Name	Version
Definition	Manages version information of the system. This version information uses Version class defined in .NET Framework.	

Table218.Definition of Version Information

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	Major	Major			O	Int32	Major version number	1.10	R/W		R	
2	Minor	Minor			O	Int32	Minor version number	1.10	R/W		R	
3	Build	Build			O	Int32	Hotfix version number	1.10	R/W		R	
4	Revision	Revision			O	Int32	Revision version number * unused for this system	1.10	R/W		R	
5	Major Revision	MajorRevision			O	Int16	Upper 16 bits of Revision * unused for this system	1.10	R		R	
6	Minor Revision	MinorRevision			O	Int16	Lower 16 bits of Revision * unused for this system	1.10	R		R	

2.3.18 LDAP Server Setting

Table219. LDAP Server Setting

Entity Name	Logical Name	LDAP Server Setting
	Physical Name	LdapServer_Setting
Definition	Holds LDAP Server connection and registration information. Setting values for the AdminTool [LDAP Authentication Setting] window.	

Table220. Definition of LDAP Server Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access			
									AdminTool	Receiver	Web Server	Exporter
1	LDAP Authentication	enableldap			O	Bool	true: Enabled false: Disabled	1.10	R/W		R	
2	Automatic LDAP Registration	enableentry			O	Bool	true: Enabled false: Disabled	1.10	R/W		R	
3	Server Name	servername	O		O	String	* For Ver 1.10, fixed to "LDAPServer"	1.10	R/W		R	
4	IP Address / Host Name	ipaddress			O	String		1.10	R/W		R	
5	Port Number	port			O	Int32		1.10	R/W		R	
6	LDAPS	ldaps			O	Bool	*Unused	1.10	R/W		R	
7	Domain	domain			O	String		1.10	R/W		R	
8	Account	account			O	String		1.10	R/W		R	
9	Password	password			O	String	To be encrypted when saved	1.10	R/W		R	
10	Login ID Attribute	ldapid			O	String		1.10	R/W		R	
11	Search Scope	basedn			O	String		1.10	R/W		R	
12	LDAP Filter	ldapfilter				String		1.10	R/W		R	
13	Language	lang				String	Used for switching display jp: Japanese en: English cn: Chinese local: Local language	1.10	R/W		R	
14	Role ID	roleids			O	Array	Assigned to users to control their authorities. Array of role names (String)	1.10	R/W		R	
15	Calendar Period	calendarspan			O	Array	Refer to "**2.3.19.1 Calendar Period Setting**"	1.10	R/W		R	
16	Format Version	formatversion			O	Int32	Version of function	1.10	R/W		R	

2.3.19 Calendar Period

Table221. Calendar Period

Entity Name	**Logical Name**	Calendar Period
	Physical Name	user_calendar_span
Definition	Manages the calendar period of each function for each user.	

Table222. Definition of Calendar Period

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	User ID	userid			O	Int32	Start from 1	1.0	W		
2	Calendar Period Setting	calendarspan			O	Array	Refer to "**2.3.19.1 Calendar Period Setting**"	1.0	R/W		R
3	Format Version	formatversion			O	Int32	Version of function	1.0	W		

2.3.19.1 Calendar Period Setting

Table223. Calendar Period Setting

Array Name	**Logical Name**	Calendar Period Setting
	Physical Name	Calendarspan
Definition	Sets the calendar period (initial display period / maximum period) for each function (display).	

Table224. Definition of Calendar Period

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Screen Name	pagename			O	String	"bargraph" "piegraph" "productresult" "alarm"	1.0	R/W		R
2	Initial Display Period	initspan			O	Int32	Unit: day	1.0	R/W		R
3	Maximum Period	maxspan			O	Int32	Unit: day	1.0	R/W		R

2.3.20 Facility Layout Arrangement Information Role Setting

Table225. Facility Layout Arrangement Information Role Setting

Entity Name	**Logical Name**	Facility Layout Arrangement Information Role Setting
	Physical Name	role_layout
Definition	Manages role setting of facility layout arrangement.	

Table226. Definition of Facility Layout Arrangement Information Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Role ID	roleid			O	Int32		1.7	R/W		
2	Calendar Period Setting	auth_list			O	Array	Array (func_auth) Refer to "2.3.20.1 Function Role Setting"	1.7	R/W		R
3	Format Version	formatversion			O	Int32	Version of function	1.7	R/W		

2.3.20.1 Function Role Setting

Table227. Function Role Setting

Array Name	**Logical Name**	Function Role Setting
	Physical Name	auth_list
Definition	Assigns roles for displaying the facility layout arrangement.	

Table228. Definition of Function Role Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Function Name	functionname			O	String	"facilitylayoutarrangement"	1.7	R/W		R
2	Facility Layout Arrangement Name	pagename			O	String		1.7	R/W		R
3	Role	auth_level			O	Int32	0: No authority 1: ReadOnly (Default) 2: Read/Write	1.7	R/W		R

2.3.21 Facility Layout Arrangement Setting

Table229. Facility Layout Arrangement Setting

Entity Name	**Logical Name**	Facility Layout Arrangement Setting
	Physical Name	facilitylayoutarrangement_setting
Definition	Manages facility layout arrangement setting.	

Table230. Definition of Facility Layout Arrangement Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility Layout Arrangement ID	id			O	Int32	Facility Layout Arrangement ID Value: 1 to 10	1.7	R/W		R
2	Facility Layout Arrangement Name	name			O	String		1.7	R/W		R
3	Number of Rows	row			O	Int32	Set number of rows Value: 1 to 5 (rows)	1.7	R/W		R
4	Number of Columns	column			O	Int32	Set number of columns Value: 1 to 5 (columns)	1.7	R/W		R
5	BOX Setting	boxlist			O	Array	Array (boxinfo) Refer to "**2.3.21.1 BOX Setting**"	1.7	R/W		R
6	Format Version	formatversion			O	Int32	Version of function	1.7	R/W		

2.3.21.1 BOX Setting

Table231. BOX Setting

Array Name	Logical Name	BOX Setting
	Physical Name	boxinfo
Definition	Manages BOX setting on facility layout arrangement.	

Table232. Definition of BOX Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Row Number	row_index			O	Int32	Set row number Value: 0 to 4	1.7	R/W		R
2	Column Number	column_index			O	Int32	Set column number Value: 0 to 4	1.7	R/W		R
3	Facility ID	facilityid			O	Int64	Set "Facility ID" of facility master	1.7	R/W		R
4	Layout Name	layoutname			O	String	Layout name set for each facility	1.7	R/W		R

2.3.22 Facility Synchronization Setting

Table233. Facility Synchronization Setting

Array Name	**Logical Name**	Facility Synchronization Setting
	Physical Name	facilitysyncro_setting
Definition	Manages facility synchronization setting.	

Table234. Definition of Facility Synchronization Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Facility ID	facilityid		O	O	Int64	Set "Facility ID" of facility master	1.7	R/W		
2	Layout Setting	layoutlist			O	Array	Array (layoutinfo) Refer to "**2.3.22.1 Layout Setting**"	1.7	R/W		
3	Format Version	formatversion			O	Int32	Version of function	1.7	R/W		

2.3.22.1 Layout Setting

Table235. Layout Setting

Array Name	**Logical Name**	Layout Setting
	Physical Name	layoutinfo
Definition	Manages layout setting of each facility.	

Table236. Definition of Layout Setting

#	Attribute (Logical)	Attribute (Physical)	PK	FK	NN	Data Type	Definition	Ver.	Application Access		
									AdminTool	Importer	Web Server
1	Layout Name	layoutname			O	String		1.7	R/W		
2	Path of Background Image	backimagepat h			O	String	Path of background image file (NOTE)	1.7	R/W		

NOTE
This is the relative path from Integration Server AdminTool installation folder.
The default Integration Server AdminTool installation folder is shown below.
C:\FANUC\MT-LINKi-IntegrationServer\AdminTool

2.4 STATUS

(1) Status information of each collection is kept in the image of the following:

{
Signalname : [Signal Name]
  value: [Signal Value],
  filter: [Converted Signal Value (Score Value)]
  judge: [Result],
}

For examples of the stored data, refer to "**3.2 EQUIPMENT SIGNAL CHANGE POINT HISTORY**".

For signal names, refer to the "LIST OF FanucCNC SIGNALS" and "LIST OF FanucRobot SIGNALS" subsections of ""FANUC MT-LINK*i* AdminTool OPERATOR'S MANUAL."

[Simple Threshold Checker] is used as the outlier detection type. Filter Value is set to "null" when filtering is not specified.

Judge is set with the results of the threshold (0: success, 1: alert, 2: failure) specified by outlier detection. Determination Value is set to "null" when outlier detection type is not specified. (Version 2.0 does not support outlier detection.)

The result of the threshold value displays "Warning: w, Error: E" as the result for the items JudgeUpper/JudgeLower, as stated in the "FANUC MT-LINK*i* WebUI OPERATOR'S MANUAL, 3.15.1 Signal History Screen (Transited from Main Menu). ".

Outlier detection type: Threshold value of Simple Threshold Checker

Range	Error	Warning
Or more (JudgeUpper)	290.0	200.0
Or less (JudgeLower)	-290.0	-200.0

Example of obtained/stored data (Number of smoothing data: 1)

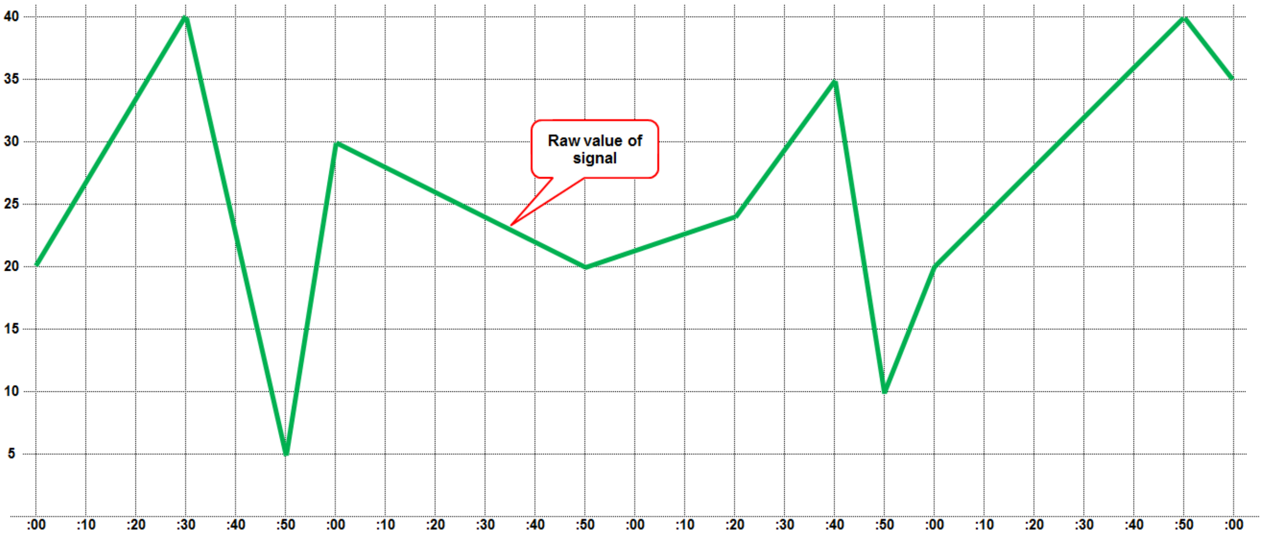
Date/time	Value	Filter	Judge	TypeID	timespan
2016-11-07T08:24:44.500Z	286.997	286.997	1	10001	0.5
2016-11-07T08:24:45.500Z	288.906	288.906	1	10001	0.5
2016-11-07T08:24:46.000Z	289.868	289.868	1	10001	0.5
2016-11-07T08:24:46.500Z	290.846	290.846	2	10001	0.5

Example of obtained/stored data (Number of smoothing data: 3)

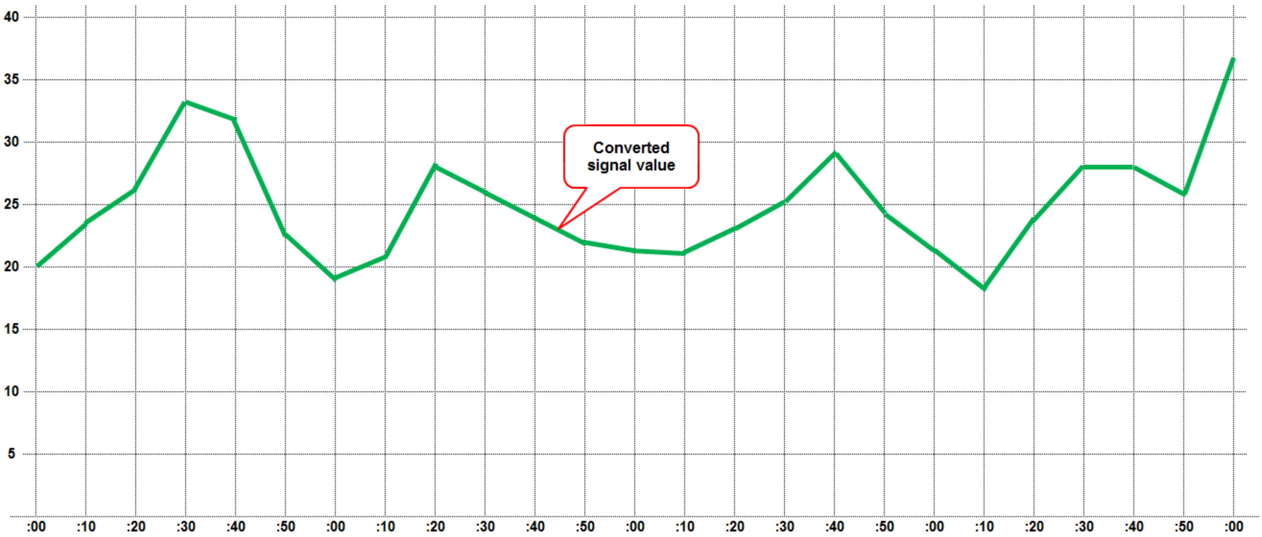
Date/time	Value	Filter	Judge	TypeID	timespan
2016-11-08T08:55:44.500Z	100.0	93.990000	0	10001	0.5
2016-11-08T08:55:45.500Z	100.0	98.129	0	10001	0.5
2016-11-08T08:55:46.000Z	100.0	100.0	0	10001	0.5
2016-11-08T08:55:46.500Z	82.667	94.222333	0	10001	0.5

As shown above, the value changes depending on the parameter specified as the outlier detection type.
Value of the Filter is the average value of the sum of its current Value and the preceding values counting up to the number of smoothing data specified.

When number of smoothing data pieces is set at [1]



When number of smoothing data pieces is set at [3]



2.5 SETTINGS AND RESULTS

(1) When settings are changed, the changed items are copied to the corresponding result collections if needed.

2.6 DELETION

(1) An Equipment Setting item, or a Status Setting item under an Equipment Setting item has a Deletion Flag attribute that indicates whether the item is deleted.
(2) When filtering equipment or signals, a search on monitoring screens will be done only with active candidates, and a search on history screens will be done with all candidates including deleted items.

2.7 EQUIPMENT AND GROUPS

(1) A Group Setting item has an Equipment Name array of equipment which a group consists of, and the Equipment Setting item has a Group Name attribute of the group which the equipment belongs to.

2.8 VIRTUAL ITEMS

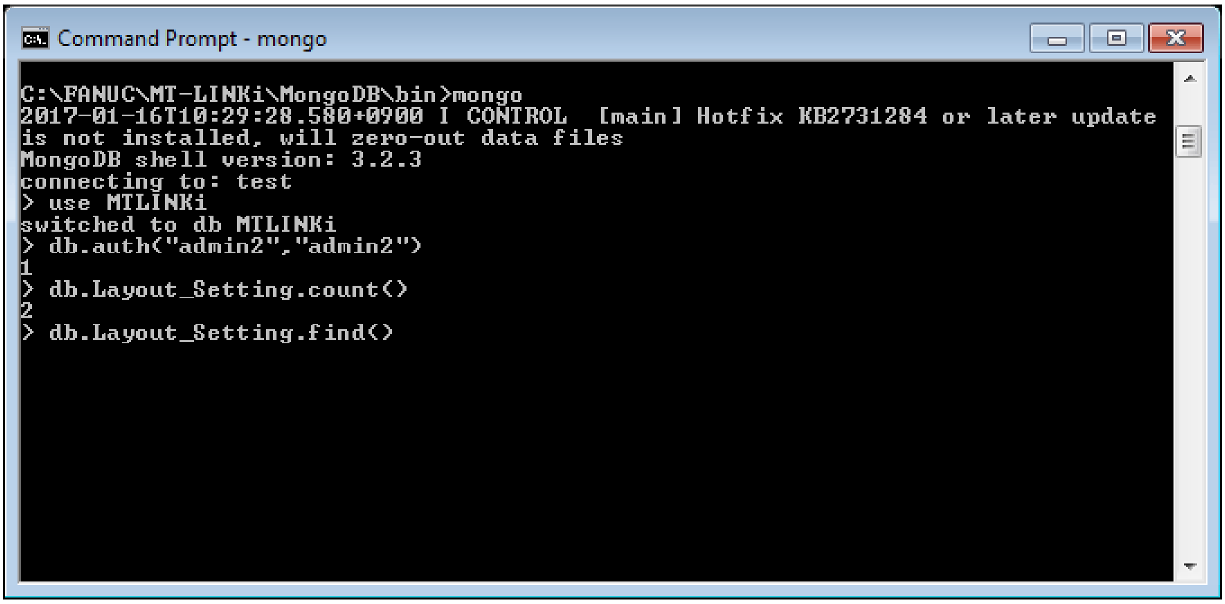
(1) If the start or the end time to view history information is outranged of the history period, the start or end time will be adjusted to review the maximum history information if needed.
(2) All the information subject to the language is stored in English, and the information expressed in a non-English language is obtained virtually when language setting is changed from English.

3 COLLECTION BROWSING

Following is an example showing how to browse the database by using the command prompt.

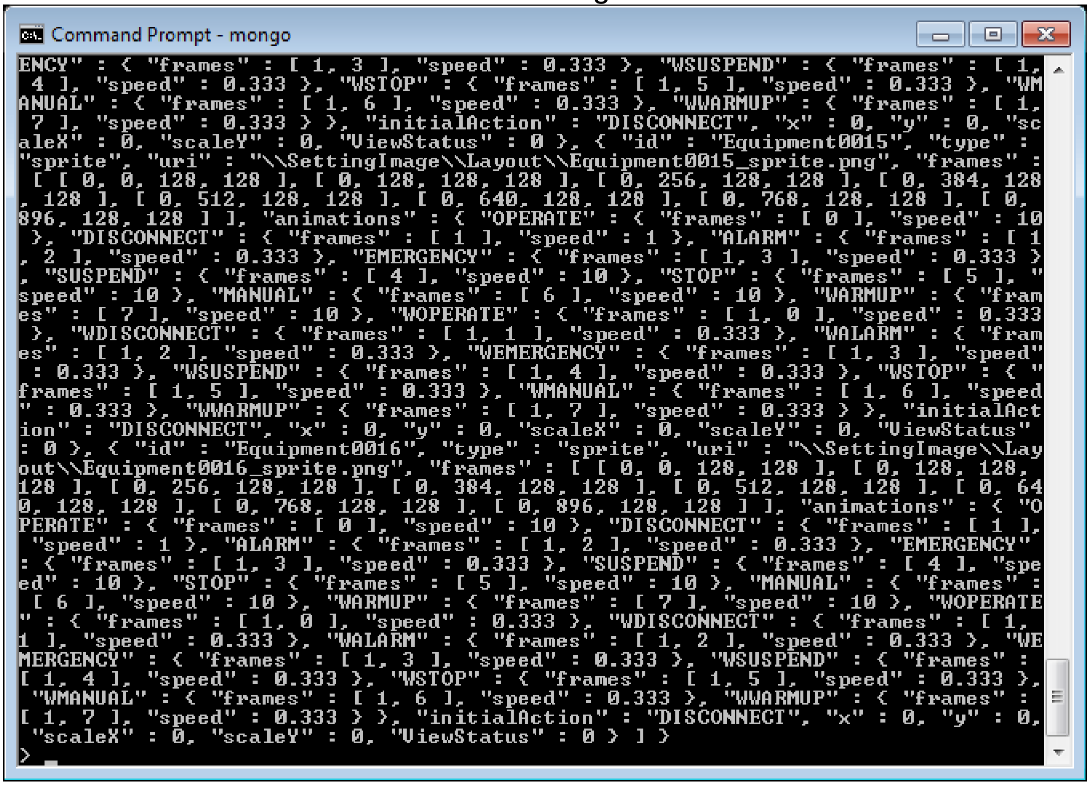
Connect to the DB
(1) Launch [Mongo] from the command prompt.
(2) Browse the stored database. [use MTLINKi]
(3) For the user authentication, enter Username and Password. [db.auth ("Username", "Password")]
Enter steps (1) to (3) altogether as [Mongo MTLINKi -u Username -p Password] to connect to the DB.

Browse the contents of the Collection (Layout_Setting)
(4) Example for browsing the number of collections saved. [db.Layout_Setting.count()]
(5) Example for browsing the collection data. [db.Layout_Setting.find()]



NOTE

When executing [find()] in the command prompt, all the contents may not be shown in the command prompt screen even after scrolling. An example is shown below. It is recommended to store in files when browsing the contents of the DB.


Command Prompt - mongo

```
ENCY" : { "frames" : [ 1, 3 ], "speed" : 0.333 }, "WSUSPEND" : { "frames" : [ 1,
 4 ], "speed" : 0.333 }, "WSTOP" : { "frames" : [ 1, 5 ], "speed" : 0.333 }, "WM
ANUAL" : { "frames" : [ 1, 6 ], "speed" : 0.333 }, "WWARMUP" : { "frames" : [ 1,
 7 ], "speed" : 0.333 } }, "initialAction" : "DISCONNECT", "x" : 0, "y" : 0, "sc
aleX" : 0, "scaleY" : 0, "ViewStatus" : 0 }, { "id" : "Equipment0015", "type" :
"sprite", "uri" : "\\SettingImage\\Layout\\Equipment0015_sprite.png", "frames" :
 [ [ 0, 0, 128, 128 ], [ 0, 128, 128, 128 ], [ 0, 256, 128, 128 ], [ 0, 384, 128
, 128 ], [ 0, 512, 128, 128 ], [ 0, 640, 128, 128 ], [ 0, 768, 128, 128 ], [ 0,
896, 128, 128 ] ], "animations" : { "OPERATE" : { "frames" : [ 0 ], "speed" : 10
 }, "DISCONNECT" : { "frames" : [ 1 ], "speed" : 1 }, "ALARM" : { "frames" : [ 1
, 2 ], "speed" : 0.333 }, "EMERGENCY" : { "frames" : [ 1, 3 ], "speed" : 0.333 }
, "SUSPEND" : { "frames" : [ 4 ], "speed" : 10 }, "STOP" : { "frames" : [ 5 ], "
speed" : 10 }, "MANUAL" : { "frames" : [ 6 ], "speed" : 10 }, "WARMUP" : { "fram
es" : [ 7 ], "speed" : 10 }, "WOPERATE" : { "frames" : [ 1, 0 ], "speed" : 0.333
 }, "WDISCONNECT" : { "frames" : [ 1, 1 ], "speed" : 0.333 }, "WALARM" : { "fram
es" : [ 1, 2 ], "speed" : 0.333 }, "WEMERGENCY" : { "frames" : [ 1, 3 ], "speed"
 : 0.333 }, "WSUSPEND" : { "frames" : [ 1, 4 ], "speed" : 0.333 }, "WSTOP" : { "
frames" : [ 1, 5 ], "speed" : 0.333 }, "WMANUAL" : { "frames" : [ 1, 6 ], "speed
" : 0.333 }, "WWARMUP" : { "frames" : [ 1, 7 ], "speed" : 0.333 } }, "initialAct
ion" : "DISCONNECT", "x" : 0, "y" : 0, "scaleX" : 0, "scaleY" : 0, "ViewStatus"
: 0 }, { "id" : "Equipment0016", "type" : "sprite", "uri" : "\\SettingImage\\Lay
out\\Equipment0016_sprite.png", "frames" : [ [ 0, 0, 128, 128 ], [ 0, 128, 128,
128 ], [ 0, 256, 128, 128 ], [ 0, 384, 128, 128 ], [ 0, 512, 128, 128 ], [ 0, 64
0, 128, 128 ], [ 0, 768, 128, 128 ], [ 0, 896, 128, 128 ] ], "animations" : { "O
PERATE" : { "frames" : [ 0 ], "speed" : 10 }, "DISCONNECT" : { "frames" : [ 1 ],
 "speed" : 1 }, "ALARM" : { "frames" : [ 1, 2 ], "speed" : 0.333 }, "EMERGENCY"
: { "frames" : [ 1, 3 ], "speed" : 0.333 }, "SUSPEND" : { "frames" : [ 4 ], "spe
ed" : 10 }, "STOP" : { "frames" : [ 5 ], "speed" : 10 }, "MANUAL" : { "frames" :
 [ 6 ], "speed" : 10 }, "WARMUP" : { "frames" : [ 7 ], "speed" : 10 }, "WOPERATE
" : { "frames" : [ 1, 0 ], "speed" : 0.333 }, "WDISCONNECT" : { "frames" : [ 1,
1 ], "speed" : 0.333 }, "WALARM" : { "frames" : [ 1, 2 ], "speed" : 0.333 }, "WE
MERGENCY" : { "frames" : [ 1, 3 ], "speed" : 0.333 }, "WSUSPEND" : { "frames" :
[ 1, 4 ], "speed" : 0.333 }, "WSTOP" : { "frames" : [ 1, 5 ], "speed" : 0.333 },
 "WMANUAL" : { "frames" : [ 1, 6 ], "speed" : 0.333 }, "WWARMUP" : { "frames" :
[ 1, 7 ], "speed" : 0.333 } }, "initialAction" : "DISCONNECT", "x" : 0, "y" : 0,
 "scaleX" : 0, "scaleY" : 0, "ViewStatus" : 0 } ] }
>
```

< Use the command prompt to store the file >

Store the data of the DB in a file.

Create a batch file and store it in a file using the redirect.

Example 1:

Create a text file for loading data: dbread.txt

```
db.L1Signal_Pool.find()
it
exit
```

Create a batch file to extract the data: dbread.bat

```
path c:\mongodb\bin
mongo MTLINKi –u Username –p Password < dbread.txt > dbread.log
```

Browse the loaded data (dbread.log).

```
C:\Users\user>path C:\FANUC\MT-LINKi\MongoDB\bin

C:\Users\user>type dbread.txt
db. L1Signal_Pool.find ()
it
exit

C:\Users\user>mongo MTLINKi -u MTLINKi -p adminUser < dbread.txt > dbread.log

C:\Users\user>type dbread.log
2017-01-20T14:11:11.784+0900 I CONTROL  [main] Hotfix KB2731284 or later update
is not installed, will zero-out data files
MongoDB shell version: 3.2.3
connecting to: MTLINKi
{ "_id" : ObjectId("57aa8179eba2831838e5bdb4"), "L1Name" : "Equipment0001", "upd
atedate" : ISODate("2016-08-10T01:18:51Z"), "enddate" : ISODate("2016-08-10T01:2
0:58Z"), "timespan" : 127.00032399999999, "signalname" : "Disconnect_Machine001"
, "value" : true, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : nu
ll, "Warning" : null }
{ "_id" : ObjectId("57aa8179eba2831838e5bdb5"), "L1Name" : "Equipment0001", "upd
atedate" : ISODate("2016-08-10T01:18:51Z"), "enddate" : ISODate("2016-08-10T01:2
0:58Z"), "timespan" : 127.00032399999999, "signalname" : "SigOP_path1_Machine001
", "value" : null, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : n
ull, "Warning" : null }
{ "_id" : ObjectId("57aa8179eba2831838e5bdb6"), "L1Name" : "Equipment0001", "upd
atedate" : ISODate("2016-08-10T01:18:51Z"), "enddate" : ISODate("2016-08-10T01:2
0:58Z"), "timespan" : 127.00032399999999, "signalname" : "SigSTL_path1_Machine00
1", "value" : null, "filter" : null, "TypeID" : null, "Judge" : null, "Error" :
null, "Warning" : null }
{ "_id" : ObjectId("57aa8179eba2831838e5bdb7"), "L1Name" : "Equipment0001", "upd
atedate" : ISODate("2016-08-10T01:18:51Z"), "enddate" : ISODate("2016-08-10T01:2
0:58Z"), "timespan" : 127.00032399999999, "signalname" : "SigSPL_path1_Machine00
1", "value" : null, "filter" : null, "TypeID" : null, "Judge" : null, "Error" :
null, "Warning" : null }
{ "_id" : ObjectId("57aa8179eba2831838e5bdb8"), "L1Name" : "Equipment0001", "upd
atedate" : ISODate("2016-08-10T01:18:51Z"), "enddate" : ISODate("2016-08-10T01:2
```

Example 2:

Output a DB collection (Example: ProductResult_History Collection in MTLINK*i* database) to a JSON file (mtlinki_ProductResult_History00.json) by using **mongoexport.exe**.

Browse the exported text data (mtlinki_ProductResult_History00.json).

> **NOTE**
> With 1.7 million history records, for example, it takes some 20 minutes to output and the size of the output file may exceed 2 gigabytes. Usually, such a large file cannot be browsed with a text editor.
> In such a case of a huge amount of data as the example shown above, use two options (/limit and /skip) to divide the output file into smaller files with a size of around 0.1 million records.

mongoexport /d MTLINKi **/u Username /p Password** /c ProductResult_History /o

mongoexport /d MTLINKi **/u Username /p Password** /c ProductResult_History /o mtlinki_ProductResult_History01.json /limit 100000 /skip 100000

Example when 3,905 records are divided by 3,000 records

```
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\user>path C:\FANUC\MT-LINKi\MongoDB\bin

C:\Users\user>mongoexport /d MTLINKi  /u admin2 /p admin2 /c ProductResult_History /o mtlinki_ProductResult_History00.json /limit 100000
2017-01-16T11:28:01.495+0900    connected to: localhost
2017-01-16T11:28:01.662+0900    exported 3905 records

C:\Users\user>mongoexport /d MTLINKi  /u admin2 /p admin2 /c ProductResult_History /o mtlinki_ProductResult_History01.json /limit 3000
2017-01-16T11:32:08.285+0900    connected to: localhost
2017-01-16T11:32:08.409+0900    exported 3000 records

C:\Users\user>mongoexport /d MTLINKi  /u admin2 /p admin2 /c ProductResult_History /o mtlinki_ProductResult_History01.json /limit 3000 /skip 3000
2017-01-16T11:32:25.034+0900    connected to: localhost
2017-01-16T11:32:25.158+0900    exported 905 records

C:\Users\user>dir mtlink*
 Volume in drive C has no label.
 Volume Serial Number is 4467-2076

 Directory of C:\Users\user

01/16/2017  11:28         1,068,375 mtlinki_ProductResult_History00.json
01/16/2017  11:32           247,883 mtlinki_ProductResult_History01.json
               2 File(s)      1,316,258 bytes
               0 Dir(s)  249,052,016,640 bytes free

C:\Users\user>
```

3.1 MANAGEMENT OF PRODUCTION RESULT

Below is an example of loading production results from history data (ProductResult_History) in the DB.

In this example, the production results of Equipment0001 from 2016/2/29 16:49:10.0 are browsed and the information such as the following 2 items can be browsed.

Equipment Name: Equipment0001, Date and Time: 2016/2/29 16:49:27.0, Number of Results: 1

{"L1Name":"**Equipment0001**","_id":{"$oid":"56d3f807e878301da0db915f"},"enddate":{"$date":"**2016-02-29T07:49:27.000Z**"}, "productname":"SAMPLE2","productresult":**1**,"productresult_accumulate":**2983**,"resultflag":true,"timespan":18.0,"update date":{"$date":"**2016-02-29T07:49:09.000Z**"}}

Equipment Name: Equipment0001, Date and Time: 2016/2/29 16:49:27.0 - 16:49:45.0, Number of Results: 1

{"L1Name":"**Equipment0001**","_id":{"$oid":"56d3f819e878301da0db9831"},"enddate":{"$date":"**2016-02-29T07:49:45.000Z**"} ,"productname":"SAMPLE2","productresult":**1**,"productresult_accumulate":**2984**,"resultflag":true,"timespan":18.0,"update date":{"$date":"**2016-02-29T07:49:27.000Z**"}}

As shown above, production result can be managed through a time-series management function, Equipment History. In the case of the example above, the computation of Production Result generates one ("1") as the number of production results from 16:49:27 to 16:49:45.0 on 2016/2/29.

Date: "updatedate" is set in GMT offset for the time zone specified on Server PC.

3.2 EQUIPMENT SIGNAL CHANGE POINT HISTORY

Following is an example of getting from DB stored data of Equipment Signal Change Point History (L1Signal_Pool), the simple assign status: automatic operation/Integration values of operating time of Equipment: Equipment0001, Date/time: 2016/02/29 17:25:40.0 and after.

Equipment Name: Equipment0001, Date and Time: 2016/2/29 17:25:53.0, Automatic operation signal: **false**


```
{
    "_id" : ObjectId("5876d493e878300c48b3c469"),
    "L1Name" : "Equipment0001", "updatedate" : ISODate("2016-02-29T08:25:53.000+0000"),
    "enddate" : ISODate("2016-02-29T08:26:41.000+0000"), "timespan" : 108.0,
    "signalname" : "SigOP_path1_NewMachine001",
    "value" : false, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : null, "Warning" : null
}
```


Equipment Name: Equipment0001, Date and Time: 2016/2/29 17:25:46.0, Integration values of operating time: **425454.0**


```
{
    "_id" : ObjectId("5876d36fe878300c48b3c41e"),
    "L1Name" : "Equipment0001", "updatedate" : ISODate("2016-02-29T08:25:46.000+0000"),
    "enddate" : ISODate("2016-02-29T08:26:41.000+0000"), "timespan" : 55.0,
    "signalname" : "RunTime_path1_NewMachine001",
    "value" : 425454.0, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : null, "Warning" : null
}
```


Equipment Name: Equipment0001, Date and Time: 2016/2/29 17:26:41.0, Automatic operation signal: **true**


```
{
    "_id" : ObjectId("5876d4ace878300c48b3c51c"),
    "L1Name" : "Equipment0001", "updatedate" : ISODate("2016-02-29T08:26:41.000+0000"),
    "enddate" : ISODate("2016-02-29T08:27:06.000+0000"), "timespan" : 25.0,
    "signalname" : "SigOP_path1_NewMachine001",
    "value" : true, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : null, "Warning" : null
}
```


Equipment Name: Equipment0001, Date and Time: 2016/2/29 17:27:15.0, Integration values of operating time: **425598.0**


```
{
    "_id" : ObjectId("5876d36fe878300c48b3c41e"),
    "L1Name" : "Equipment0001", "updatedate" : ISODate("2016-02-29T08:27:15.000+0000"),
    "enddate" : ISODate("2016-02-29T08:27:18.000+0000"), "timespan" : 3.0,
    "signalname" : "RunTime_path1_NewMachine001",
    "value" : 425598.0, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : null, "Warning" : null
}
```


Equipment Name: Equipment0001, Date and Time: 2016/2/29 18:05:59.0 Integration values of operating time: **427922.0**


```
{
    "_id" : ObjectId("5876d36fe878300c48b3c41e"),
    "L1Name" : "Equipment0001", "updatedate" : ISODate("2016-02-29T09:05:59.000+0000"),
    "enddate" : ISODate("2016-02-29T09:06:00.000+0000"), "timespan" : 1.0,
    "signalname" : "RunTime_path1_NewMachine001",
    "value" : 427922.0, "filter" : null, "TypeID" : null, "Judge" : null, "Error" : null, "Warning" : null
}
```

NOTE

1 Signal information: Signal information which is not defined in Equipment Setting ([2.2.7.1 Status Setting]) is not stored.
It does not mean that the unsaved data has no value. In the preceding example, if [SigOP_path1] is undefined, the data will be stored without the step of ["SigOP_path1":{…},].

2 The information about signal names is described in the "LIST OF FanucCNC SIGNALS" and "LIST OF FanucRobot SIGNALS" subsections of ""FANUC MT-LINK*i* AdminTool OPERATOR'S MANUAL."

Example of the Contents of Simple Assign Status: Automatic Operation/Integration Values of Operating Time in the Machine Setting

{"CncMaxPath" : 1 , "CollectorIndex" : 1 , "Connect" : true , "FormatVersion" : 4 , "Index" : 0 , "L0CnName" : "机器 001" , "L0EnName" : "Machine001" , "L0JpName" : "機械 004" , "L0Name" : "Machine001" , "MachineType" : 0, "MacroCaputreDefine" : [{ "CNCPath" : 1 , "CaptureMacro" : 501 , "CaputureCount" : 1 , "CnID" : "ID0001" , "EnID" : "ID0001" , "FormatVersion" : 1 , "ID" : "ID0001" , "JpID" : "ID0001 日本" , "TriggerMacro" : 500 } , { "CNCPath" : 1 , "CaptureMacro" : 501 , "CaputureCount" : 1 , "CnID" : "ID0002" , "EnID" : "ID0002" , "FormatVersion" : 1 , "ID" : "ID0002" , "JpID" : "ID0002 日本語" , "TriggerMacro" : 502 }] , "NetworkSetting" : { "FormatVersion" : 1 , "IpAddress" : "10.15.9.15" , "Port" : 8193 , "TimeOut" : 30 , "datatype" : 0 } , "OperationHistorySchedule" : { "ExcludeDayList" : [ ] , "FormatVersion" : 1 , "ScheduleList" : [ ] } , "SamplingCycle" : 500 , "Signal_Setting" : [{ "Filter_Setting" : null , "FormatVersion" : 2 , "MessageSetting" : null , "SignalCnName" : "自动运转中信号" , "SignalEnName" : "Automatic operation signal" , "SignalJpName" : "自動運転中信号" , "SignalLabel" : { "Adrs" : "" , "Bit" : -1 , "CncPath" : 1 , "DataType" : -1 , "Enable" : true , "FormatVersion" : 1 , "LabelName" : "**SigOP**" , "LabelType" : 0 , "Num" : -1 , "dcLabelFANUCFormatVersion" : 2 , "isCustom" : false } , "SignalName" : "**SigOP_path1**" , "SignalType" : 0 , "Type" : "Bool" } , : { "Filter_Setting" : null , "FormatVersion" : 2 , "MessageSetting" : null , "SignalCnName" : "运行时间" , "SignalEnName" : "Operation time" , "SignalJpName" : "運転時間" , "SignalLabel" : { "Adrs" : "" , "Bit" : -1 , "CncPath" : 1 , "DataType" : -1 , "Enable" : true , "FormatVersion" : 1 , "LabelName" : "**RunTime**" , "LabelType" : 0 , "Num" : -1 , "dcLabelFANUCFormatVersion" : 2 , "isCustom" : false } , "SignalName" : "**RunTime_path1**" , "SignalType" : 0 , "Type" : "Double" } ,{ "Filter_Setting" : null , "FormatVersion" : 2 , "MessageSetting" : null , "SignalCnName" : "MacroVar_1_path1" , "SignalEnName" : "MacroVar_1_path1" , "SignalJpName " : "MacroVar_1_path1" , "SignalLabel" : { "Adrs" : "" , "Bit" : -1 , "CncPath" : 1 , "DataType" : -1 , "Enable" : true , "FormatVersion" : 1 , "LabelName" : "MacroVar" , "LabelType" : 0 , "Num" : 1 , "dcLabelFANUCFormatVersion" : 2 , "isCustom" : true } , "SignalName": "MacroVar_1_path1" , "SignalType" : 0 , "Type" : "Double" } , { "Filter_Setting" : null , "FormatVersion" : 2 , "MessageSetting" : null , "SignalCnName" : "PcodeCon_2_path1" , "SignalEnName" : "PcodeCon_2_path1" , "SignalJpName" : "PcodeCon_2_path1" , "SignalLabel" : { "Adrs" : "" , "Bit" : -1 , "CncPath" : 1 , " DataType" : 100 , "Enable" : true , "FormatVersion" : 1 , "LabelName" : "PcodeVar" , "LabelType" : 0 , "Num" : 2 , "dcLabelFANUCFormatVersion" : 2 , "isCustom" : true } , "SignalName": "PcodeCon_2_path1" , "SignalType" : 0, "Type" : "Double" }] , "SpindleUnit" : 0 , "_id" : { "$oid" : "56d3febce878301994200889" } }

Example of the Contents of Simple Assign Status: Automatic Operation/Integration Values of Operating Time in the Machine Setting

{ "CycleTime" : 500 , "DBStored" : true , "FormatVersion" : 5 , "ImageAlarm" : "\\SettingImage\\L1\\Equipment0001_003.png", "ImageConnect" : "\\SettingImage\\L1\\Equipment0001_001.png", "ImageDisconnect" : "\\SettingImage\\L1\\Equipment0001_002.png", "ImageEmergency" : "\\SettingImage\\L1\\Equipment0001_004.png", "ImageManual" : "\\SettingImage\\L1\\Equipment0001_007.png", "ImageStop" : "\\SettingImage\\L1\\Equipment0001_006.png", "ImageSuspend" : "\\SettingImage\\L1\\Equipment0001_005.png", "ImageWarmup" : "\\SettingImage\\L1\\Equipment0001_008.png", "L1CnName" : "设备 0001" , "L1EnName" : "Equipment0001" , "L1JpName" : "設備 1　a " , "L1Name" : "Equipment0001" , "L1Names" : [ ] , "LoadedInitialName" : "Equipment0001" , "TagNames" : ["Group0001" , "Group0012" , "Group0013"] , "_id" : { "$oid" : "56d3febce87830199420088b" } , "custom_status" : [ ] , "deleted_status" : [ ] , "deletedate" : "2016-02-29T08:18:04.5842799Z" , "deleteflag" : 0 , "mapped_status" : [{ "DBStored" : true , "FormatVersion" : 4 , "Formula" : "Machine001_Disconnect" , "L0Name" : "Machine001" , "LoadedInitialStatusName" : "Disconnect_Machine001" , "StatusCnName" : "Disconnect Machine001" , "StatusEnName" : "Disconnect Machine001" , "StatusJpName" : "Disconnect Machine001" , "StatusName" : "Disconnect_Machine001" , "Type" : "Bool" , "deleteflag" : false , "representflag":false} , { "DBStored" : true , "FormatVersion" : 4 , "Formula" : "**Machine001_SigOP_path1**" , "L0Name" : "Machine001" , "LoadedInitialStatusName" : "**SigOP_path1_Machine001**" , "StatusCnName" : "自动运转中信号 Machine001 P1" , "StatusEnName" : "Automatic operation signal Machine001 P1" , "StatusJpName" : "自動運転中信号 Machine001 P1" , "StatusName" : "**SigOP_path1_Machine001**" , "Type" : "Bool" , "deleteflag" : false , "representflag" : false } , {"DBStored" : true , "FormatVersion" : 4 , "Formula" : "**Machine001_RunTime_path1**" , "L0Name" : "Machine001" , "LoadedInitialStatusName" : "**RunTime_path1_Machine001**" , "StatusCnName" : "运行时间 Machine001 P1" , "StatusEnName" : "Operation time Machine001 P1" , "StatusJpName" : "運転時間 Machine001 P1" , "StatusName" : "**RunTime_path1_Machine001**" , "Type" : "Double" , "deleteflag" : false , "representflag" : false } , { "DBStored" : true , "FormatVersion" : 4 , "Formula" : "PartsNum_path1_Machine001" , "L0Name" : "PartsNum_path1_Machine001" , "LoadedInitialStatusName" : "ProductResultNumber" , "StatusCnName" : "加工零件数" , "StatusEnName" : "ProductResultNumber" , "StatusJpName" : "加工部品数" , "StatusName" : "ProductResultNumber" , "Type" : "Int" , "deleteflag" : false , "representflag" : true }] }

4 COLLECTION REFERENCE FOR INTEGRATION SERVER

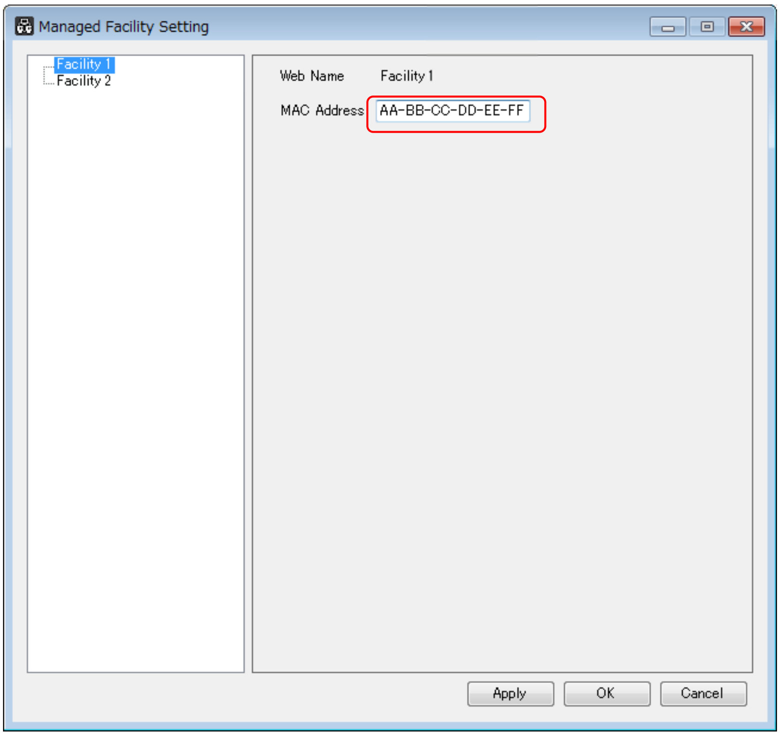
Following is an example of how to browse database data by using command prompt.

(1) Launch [Mongo –port 27020] from command prompt.
(2) Browse the stored database. [use IntegrationServer]
(3) For user authentication, enter Username and Password. [db.auth ("Username", "Password")]

That is, use the following command from command prompt:
Mongo IntegrationServer -u *username* -p *password* -port 27020

4.1 LIST OF EQUIPMENT IN FACILITY

Shown below is an example of reading equipment list by specifying MAC address of a managed facility.



Store the data in a file.
Create a batch file and store it in a file using the redirect.

Create a text file for loading data: dbread_equipment.txt

```
p=db.mt_mstr.findOne({"macaddress":"AA-BB-CC-DD-EE-FF"},{facilityid:1});
db.equipment_mstr.find({facilityid:p.facilityid});
it
exit
```

Create a batch file to extract the data: dbread_equipment.bat

```
path c:\mongodb\bin
mongo IntegrationServer –u Username –p Password –port 27020
< dbread_equipment.txt > dbread_equipment.log
```

Browse the output text data (dbread_equipment.log).

4.2 READING OPERATIONAL RESULTS IN FACILITY

Below is an example of reading operational results for a half year by specifying MAC address of a managed facility, equipment name, and equipment status.

Store the data in a file.
Create a batch file and store it in a file using the redirect.

MAC address: AA-BB-CC-DD-EE-FF
Equipment name: Equipment0001
Span of time: from Jan. 1, 2016 00:00:00.0 to Jun. 30, 2016 23:59:00.0
Status: EMERGENCY

Create a text file for loading data: dbread_operationresult.txt

```
p=db.mt_mstr.findOne({"macaddress":"00-0C-29-91-62-79"},{facilityid:1});
db.operation_result.find({ "updatedate": { $gte: ISODate("2016-01-01T00:00:00.000+0000") }, $and: [ { "updatedate": { $lt: ISODate("2016-07-01T00:00:00.000+0000") } }, { $and: [ { "facilityid": p.facilityid }, { $and: [ { "equipmentname": "Equipment01" }, { $and: [ { "status": "EMERGENCY" } ] } ] } ] } ] })
it
exit
```

Create a batch file to extract the data: dbread_operationresult.bat

```
path c:\mongodb\bin
mongo IntegrationServer –u Username –p Password –port 27020
< dbread_operationresult.txt > dbread_operationresult.log
```

Browse the output text data (dbread_operationresult.log).

5 USER AUTHENTICATION

Username and Password for the user authentication can be set on a per-database basis.

Example of browsing the collection

```
mongoexport /u Username /p Password /d MTLINKi /c ProductResult_History /o mtlinki_ProductResult_History00.json --limit 100000
```

Example of adding the collection

```
mongoimport /u Username /p Password /d MTLINKi /c ProductResult_History /o mtlinki_ProductResult_History00_P.json
```

6 ROLES AND PRIVILEGES

MT-LINK*i* / Integration Server databases support the following roles which allow applications, such as external tools, to access them.

Roles	Privileges
dbOwner	The database owner can perform any administrative action on the database. This role combines the privileges granted by the readWrite, dbAdmin, and userAdmin roles.
userAdmin	The userAdmin role provides the ability to create and modify roles and users on the current database. It also indirectly provides superuser access to either the database or, if scoped to the admin database, the cluster. Users with this role can grant any user any privilege, including themselves.
readWrite	This role provides all the privileges of the read role plus ability to modify data on all non-system collections and the system.js collection.
hostManager	This role provides the ability to monitor and manage servers.
clusterMonitor	This role provides read-only access to monitoring tools, such as the MongoDB Management Service (MMS) and Ops Manager monitoring agent.

Following are the names of MT-LINK*i* / Integration Server databases:

- MT-LINK*i* database: MTLINKi
- Integration Server database: IntegrationServer

NOTE
It is not guaranteed that any external tool can access those databases.
For the details of roles and privileges, refer to documents about MongoDB or technical information posted on the MongoDB Web site.